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Catawba Unit 1 Measurement Uncertainty Recapture (MUR) Power Uprate: Reactor Vessel Integrity and Neutron Fluence Evaluations

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EXECUTIVE SUMMARY

This report presents the reactor vessel (RV) integrity and neutron fluence evaluations for the Catawba Unit 1 measurement uncertainty recapture (MUR) power uprate to 3469 MWt. The RV integrity evaluations must be shown to meet the applicable U.S. Nuclear Regulatory Commission (NRC) requirements through the end of the licensed operating period. Catawba Unit 1 is licensed for 60 years of operation, which pertains to 54 effective full power years (EFPY) and is deemed end-of-life extension (EOLE).

Appendix A contains the credibility evaluation for the Catawba Unit 1 surveillance materials. Conclusions for the surveillance data credibility evaluation are contained in Appendix A of this report.

Appendix B contains the Emergency Response Guideline (ERG) limits classification for Catawba Unit 1. The ERG limits were developed in order to establish guidance for operator action in the event of an emergency situation, such as a pressurized thermal shock (PTS) event. Conclusions for the ERG limits evaluation are contained in Appendix B of this report.

The conclusions to the RV integrity evaluations are as follows:

EOLE Pressurized Thermal Shock

All of the Catawba Unit 1 RV materials are projected to remain below the 10 CFR 50.61 screening criteria values of 270°F for forgings, and 300°F for circumferentially oriented welds, through EOLE (54 EFPY). See Section 6 for more details.

EOLE Upper-Shelf Energy

All of the Catawba Unit 1 RV materials are projected to remain above the upper-shelf energy (USE) screening criterion value of 50 ft-lb (per 10 CFR 50, Appendix G) through EOLE (54 EFPY). See Section 7 for more details.

Applicability of Pressure-Temperature Limit Curves

The current Catawba Unit 1 pressure-temperature (P-T) limit curves are contained in Technical Specifications Figures 3.4.3-1 and 3.4.3-2. With a re-evaluation of surveillance data credibility, a recalculation of chemistry factors, the consideration of MUR power uprate fluence projections, along with consideration of all RV materials that are projected to achieve surface fluence levels of 1×10^{17} n/cm² or higher at 34 EFPY, the applicability of the current P-T limit curves decreased from 34 EFPY to 30.7 EFPY. See Section 8 for more details.

Surveillance Capsule Withdrawal Schedules

All in-vessel surveillance capsules have been removed from the Catawba Unit 1 RV. The guidelines of ASTM E185-82 are met, as required by 10 CFR 50, Appendix H, with consideration of the MUR power uprate. Ex-Vessel Neutron Dosimetry is installed in Catawba Unit 1 such that neutron fluence may be monitored during the period of extended operation in accordance with the Generic Aging Lessons Learned (GALL) Report. See Section 9 for more details.

1 METHOD DISCUSSION

Adjusted Reference Temperature

Per Regulatory Guide 1.99, Revision 2 (Reference 1), the following equations and variables are to be used for calculating Adjusted Reference Temperature (ART) values at the clad/base metal interface and at the RV 1/4-thickness (1/4T) and 3/4-thickness (3/4T) locations.

$$\mathbf{ART\ (^{\circ}F) = Initial\ RT_{NDT} + \Delta RT_{NDT} + Margin} \quad [\text{Eqn. 1}]$$

Where,

Initial RT_{NDT} (°F) = Reference temperature of the unirradiated material

$$\mathbf{Margin\ (^{\circ}F) = 2\sqrt{\sigma_I^2 + \sigma_{\Delta}^2}} \quad [\text{Eqn. 2}]$$

Where,

σ_I is the standard deviation for the Initial RT_{NDT} (note that σ_I is referred to as σ_U in 10 CFR 50.61). If the initial RT_{NDT} (reference nil-ductility transition temperature) is a measured value, σ_I is estimated from the precision of the test method; per WCAP-14040-A, Revision 4 (Reference 2), $\sigma_I = 0^{\circ}\text{F}$ when the initial RT_{NDT} is a measured value. Per 10 CFR 50.61 (Reference 3), when the initial RT_{NDT} is not a measured value and a generic mean initial RT_{NDT} value is used for welds with the welding flux types identified in 10 CFR 50.61, then $\sigma_I = 17^{\circ}\text{F}$.

σ_{Δ} is the standard deviation for ΔRT_{NDT} .

For plates and forgings:

$$\sigma_{\Delta} = 17^{\circ}\text{F when surveillance capsule data are not credible or not used*}$$

$$\sigma_{\Delta} = 8.5^{\circ}\text{F when credible surveillance capsule data are used*}$$

For welds:

$$\sigma_{\Delta} = 28^{\circ}\text{F when surveillance capsule data are not credible or not used*}$$

$$\sigma_{\Delta} = 14^{\circ}\text{F when credible surveillance capsule data are used*}$$

* σ_{Δ} not to exceed $0.5 * \Delta RT_{NDT}$ per Regulatory Guide 1.99, Revision 2

Shift in Reference Temperature (ΔRT_{NDT}) Calculations:

$$\mathbf{\Delta RT_{NDT}\ (^{\circ}F) = CF * FF} \quad [\text{Eqn. 3}]$$

Where,

CF (°F) = chemistry factor based on the copper (Cu) and nickel (Ni) weight % of the material or based on the results of surveillance capsule test data. If the weight percent of Cu and Ni is used to determine the CF, then the CF is obtained from Table 1 or Table 2 of Regulatory Guide 1.99, Revision 2. If surveillance capsule data are used to determine the CF, then the CF is determined as follows:

$$CF = \frac{\sum_{i=1}^n [A_i * f_i^{(0.28-0.10 \log f_i)}]}{\sum_{i=1}^n [f_i^{(0.56-0.20 \log f_i)}]} \quad [\text{Eqn. 4}]$$

Where:

n = The number of surveillance data points

A_i = The measured value of ΔRT_{NDT} (°F)**

f_i = fluence for each surveillance data point ($\times 10^{19}$ n/cm² (E > 1.0 MeV))

**If the surveillance weld copper and nickel content differs from that of the vessel weld, then the measured values of ΔRT_{NDT} (A_i in the preceding equation for CF) shall be adjusted by multiplying them by the ratio of the chemistry factor for the vessel weld (CF_{VW}) to that for the surveillance weld (CF_{SW}) based on the Cu and Ni content of the materials.

$$\Delta RT_{NDT} (\text{°F}) = (\text{measured } \Delta RT_{NDT}) * (CF_{VW} / CF_{SW}) \quad [\text{Eqn. 5}]$$

$$FF = \text{fluence factor} = f^{(0.28 - 0.10 * \log(f))} \quad [\text{Eqn. 6}]$$

Where,

f = Vessel inner wall surface fluence, 1/4T fluence, or 3/4T fluence [$\times 10^{19}$ n/cm² (E > 1.0 MeV)]. The neutron fluence at any depth in the vessel wall is calculated as follows:

$$f (\times 10^{19} \text{ n/cm}^2 \text{ (E > 1.0 MeV)}) = f_{\text{surf}} * e^{-0.24*(x)} \quad [\text{Eqn. 7}]$$

Where,

f_{surf} = Vessel inner wall surface fluence, $\times 10^{19}$ n/cm² (E > 1.0 MeV)

x = The depth into the vessel wall from the inner surface, inches

Upper-Shelf Energy (USE)

The predicted decrease in USE is determined as a function of fluence and copper content using either of the following:

1. Figure 2 of Regulatory Guide 1.99, Revision 2, Position 1.2, or
2. Surveillance program test results and Figure 2 of Regulatory Guide 1.99, Revision 2, Position 2.2.

Both methods require the use of the 1/4 thickness (1/4T) vessel fluence.

Reactor Vessel/Core Inlet Temperature (T_{cold})

Regulatory Guide 1.99, Revision 2, Position 1.3 identifies limitations of applicability for the calculations of reference temperature and upper-shelf energy. Nominal irradiation temperature is one of the limitations, wherein Regulatory Guide 1.99 indicates that the nominal irradiation temperature for which the procedures are valid is 550°F. Irradiation below 525°F should be considered to produce greater embrittlement, and irradiation over 590°F may be considered to produce less embrittlement.

It is concluded that Catawba Unit 1 operates within the 525°F and 590°F range. Thus, the Regulatory Guide 1.99, Revision 2 correlations are applicable.

Pressurized Thermal Shock

The PTS Rule, 10 CFR 50.61 (Reference 3), requires that for each pressurized water nuclear power reactor for which an operating license has been issued, the licensee shall have projected PTS reference temperature (RT_{PTS}) values accepted by the U.S. NRC for each RV beltline material at the end-of-life (EOL) fluence of the plant. This assessment must specify the basis for the projected value of RT_{PTS} for each vessel beltline material, including the assumptions regarding core-loading patterns, and must specify the copper and nickel contents and the fluence value used in the calculation. This assessment must be updated whenever there is a significant change in projected values of RT_{PTS} , or upon request for a change in the expiration date for operation of the facility. Changes to RT_{PTS} values are considered significant if either the previous value or the current value, or both values, exceed the screening criterion prior to the expiration of the operating license, including any renewed term, if applicable, for the plant.

Per 10 CFR 50.61 (Reference 3), the following equations and variables are to be used for calculating RT_{PTS} values at the clad/base metal interface of the vessel. RT_{PTS} is also referred to as the EOL RT_{NDT} .

$$RT_{PTS} (^{\circ}F) = IRT_{NDT} + M + \Delta RT_{NDT} \quad [Eqn. 8]$$

Where,

$$IRT_{NDT} (^{\circ}F) = RT_{NDT(u)} = \text{Initial Unirradiated } RT_{NDT} \text{ value}$$

$$M = \text{Margin } (^{\circ}F) = 2\sqrt{\sigma_U^2 + \sigma_{\Delta}^2} \quad [Eqn. 9]$$

Where,

$$\sigma_U = 0^{\circ}F \text{ when Initial } RT_{NDT} \text{ is a measured value}$$

$$\sigma_U = 17^{\circ}F \text{ when Initial } RT_{NDT} \text{ is a generic value}$$

For plates and forgings:

$\sigma_{\Delta} = 17^{\circ}\text{F}$ when surveillance capsule data are not credible or not used***

$\sigma_{\Delta} = 8.5^{\circ}\text{F}$ when credible surveillance capsule data are used***

For welds:

$\sigma_{\Delta} = 28^{\circ}\text{F}$ when surveillance capsule data are not credible or not used***

$\sigma_{\Delta} = 14^{\circ}\text{F}$ when credible surveillance capsule data are used***

*** σ_{Δ} not to exceed $0.5 \cdot \Delta\text{RT}_{\text{NDT}}$ per 10 CFR 50.61 (Reference 3)

$$\Delta\text{RT}_{\text{NDT}} (^{\circ}\text{F}) = \text{CF} \cdot \text{FF} \quad [\text{Eqn. 10}]$$

Where,

CF = chemistry factor ($^{\circ}\text{F}$) calculated generically for copper (Cu) and nickel (Ni) content based on Tables 1 and 2 in Reference 3 for welds and base metal, respectively (also referred to as Position 1.1). It can also be calculated using credible surveillance capsule data per Equation 5 of Reference 3 (also referred to as Position 2.1).

FF = fluence factor = $f^{(0.28 - 0.10 \cdot \log(f))}$, where the normalized neutron fluence at the clad/base metal interface on the inside surface of the vessel is $f = \Phi / (1.0 \times 10^{19})$. The units for Φ are n/cm^2 , $E > 1.0 \text{ MeV}$.

The RT_{PTS} screening criteria values are 270°F for plates, forgings and axial weld materials and 300°F for circumferential weld materials. All available surveillance data must be considered in the evaluation.

Surveillance Capsule Withdrawal Schedule

Per ASTM E185-82 (Reference 4), Section 4.15, the ΔRT_{NDT} or adjustment in reference temperature is “the difference in the 41 J (30 ft-lb_f) index temperatures from the average Charpy curves measured before and after irradiation.”

Per ASTM E185-82, Section 4.18, the USE level is “the average energy value for all Charpy specimens (normally three) whose test temperature is above the upper end of the transition region. For specimens tested in sets of three at each test temperature, the set having the highest average may be regarded as defining the upper-shelf energy.”

The surveillance capsule withdrawal schedule is generated based upon the guidelines specified in ASTM E185-82, Section 7.6. The minimum recommended number of surveillance capsules and their withdrawal times are identified in Table 1-1.

Table 1-1 Minimum Recommended Number of Surveillance Capsules and Their Withdrawal Schedule (Schedule in Terms of EFPY of the RV)			
Predicted Transition Temperature Shift at Vessel Inside Surface	≤ 100°F	> 100°F & ≤ 200°F	> 200°F
Minimum Number of Capsules	3	4	5
Withdrawal Sequence	EFPY		
First	6 ^(a)	3 ^(a)	1.5 ^(a)
Second	15 ^(b)	6 ^(c)	3 ^(d)
Third	EOL ^(e)	15 ^(b)	6 ^(c)
Fourth	- -	EOL ^(e)	15 ^(b)
Fifth	- -	- -	EOL ^(e)
Notes: (a) Or at the time when the accumulated neutron fluence of the capsule exceeds 5×10^{18} n/cm ² , or at the time when the highest predicted ΔRT_{NDT} of all encapsulated materials is approximately 50°F, whichever comes first (b) Or at the time when the accumulated neutron fluence of the capsule corresponds to the approximate EOL fluence at the RV inner wall location, whichever comes first (c) Or at the time when the accumulated neutron fluence of the capsule corresponds to the approximate EOL fluence at the RV 1/4T location, whichever comes first (d) Or at the time when the accumulated neutron fluence of the capsule corresponds to a value midway between that of the first and third capsules (e) Not less than once or greater than twice the peak EOL vessel fluence. This may be modified on the basis of previous tests. This capsule may be held without testing following withdrawal			

2 CALCULATED NEUTRON FLUENCE

2.1 INTRODUCTION

A discrete ordinates S_N transport analysis was performed for the Catawba Unit 1 reactor to determine the neutron radiation environment within the reactor pressure vessel. In this analysis, radiation exposure parameters were established on a plant- and fuel-cycle-specific basis. The neutron dosimetry sensor sets removed from the five previously withdrawn surveillance capsules [Z, Y, V, U, and X] and Ex-Vessel Neutron Dosimetry (EVND) capsules [A, B, C, D, E, and F] were re-analyzed using the current dosimetry evaluation methodology and updated neutron transport calculations. These dosimetry evaluations were used to validate the plant-specific neutron transport calculations applicable to Catawba Unit 1 and are described in Appendix C. These validated calculations form the basis for providing projections of the neutron exposure of the reactor pressure vessels for operating periods extending to 60 EFPY.

All of the calculations described in this section were based on nuclear cross-section data derived from ENDF/B-VI.3 and made use of the latest available calculational tools. Furthermore, the neutron transport evaluation methodologies follow the guidance of Regulatory Guide 1.190 (Reference 5). Additionally, the methods used to develop the calculated pressure vessel fluence are consistent with the NRC-approved methodology described in WCAP-16083-NP-A, Revision 0 (Reference 6) using the state-of-the-art three-dimensional parallel discrete ordinates radiation transport code, RAPTOR-M3G (Reference 7). RAPTOR-M3G is a three-dimensional (3-D) parallel discrete ordinates (S_N) radiation transport code. The methodology employed by RAPTOR-M3G is identical to the methodology employed by the TORT code, with a number of evolutionary solution enhancements resulting from the last two decades of research. RAPTOR-M3G has been designed from the ground-up as a parallel processing code, and adheres to modern best practices of software development. It has been rigorously tested against the TORT code (Reference 8) and benchmarked on an extensive set of real-world problems. The detailed benchmark for RAPTOR-M3G is described in Reference 7.

2.2 DISCRETE ORDINATES ANALYSIS

In performing the fast neutron exposure evaluations for the Catawba Unit 1 RV, a series of fuel cycle-specific forward transport calculations were carried out using RAPTOR-M3G for each operating cycle at Catawba Unit 1.

For the Catawba Unit 1 transport calculations, the models depicted in Figures 2-1 through 2-3 were utilized since, with the exception of the neutron pads, the reactor is octant symmetric. In each of these figures, a single octant is depicted showing the arrangement of neutron pads and surveillance capsules as applicable. With regard to these three geometries, it should be noted that the maximum exposure of the pressure vessel occurs in octants with the 12.5° neutron pad span where no surveillance capsules are present. Further, the surveillance capsules are located in octants with either the 20.0° or 22.5° neutron pad span.

In addition to the core, reactor internals, pressure vessel, and primary biological shield, the RAPTOR-M3G models developed for these octant geometries included explicit representations of the surveillance capsules, the pressure vessel cladding, and the insulation located external to the pressure vessel.

From a neutronic standpoint, the inclusion of the surveillance capsules and associated support structure in the analytical model is significant. Since the presence of the capsules and structure has a marked impact on the magnitude of the neutron flux as well as on the relative neutron and gamma ray spectra at dosimetry locations within the capsules, a meaningful evaluation of the radiation environment internal to the capsules can be made only when these perturbation effects are properly accounted for in the analysis.

In developing the RAPTOR-M3G analytical models of the reactor geometry shown in Figures 2-1 through 2-3, nominal design dimensions were employed for the various structural components. Likewise, water temperatures and coolant density in the reactor core and downcomer regions of the reactor were taken to be representative of full power operating conditions. These coolant temperatures were varied on a cycle-specific basis and are described in more detail later in this section. The reactor core itself was treated as a homogeneous mixture of fuel, cladding, water, and miscellaneous core structures such as fuel assembly grids, guide tubes, et cetera.

The RAPTOR-M3G geometric mesh description of the reactor models shown in Figures 2-1 through 2-3 consisted of 209 radial by 195 azimuthal by 179 vertical intervals for each neutron pad configuration. Mesh sizes were chosen to assure that proper convergence of the inner iterations was achieved on a pointwise basis. The pointwise inner iteration flux convergence criterion utilized in the RAPTOR-M3G calculations was set at a value of 0.001.

A section view of the RAPTOR-M3G model of the Catawba Unit 1 reactor in the r,z plane is shown in Figures 2-4 and 2-5. The model extends radially from the centerline of the reactor core out to a location interior to the primary biological shield and over an axial span from an elevation approximately six feet below the active fuel to approximately five feet above the active fuel. As can be seen in the figures, the stainless steel former plates located between the core baffle and core barrel regions were also explicitly included in the model. The volume fractions utilized for the fuel region, the bypass region, the downcomer region, the reactor pressure vessel insulation region, and the volume fractions for the regions above and below the active core were treated as a homogeneous mixture of composing materials. In regions containing reactor coolant, the coolant temperatures were varied on a cycle-specific basis and are described in more detail later in this section.

The data utilized for the core power distributions in the plant-specific transport analyses included cycle-dependent fuel assembly initial enrichments, burnups, and axial power distributions. This information was used to develop spatial- and energy-dependent core source distributions averaged over each individual fuel cycle. Therefore, the results from the neutron transport calculations provided data in terms of fuel cycle-averaged neutron flux, which when multiplied by the appropriate fuel cycle length, generated the incremental fast neutron exposure for each fuel cycle. In constructing these core source distributions, the energy distribution of the source was based on an appropriate fission split for uranium and plutonium isotopes based on the initial enrichment and burnup history of individual fuel assemblies. From these assembly-dependent fission splits, composite values of energy release per fission, neutron yield per fission, and fission spectrum were determined.

All of the transport calculations supporting this analysis were carried out using the RAPTOR-M3G parallel discrete ordinates code, version 2.0 (Reference 7), and the BUGLE-96 cross-section library (Reference 9). The BUGLE-96 library provides a 67-group coupled neutron-gamma ray cross-section data set produced specifically for light-water reactor (LWR) applications. In these analyses, anisotropic

scattering was treated with a P_3 Legendre expansion or higher and angular discretization was modeled with an S_8 order of angular quadrature or higher. Energy- and space-dependent core power distributions, as well as system operating temperatures, were treated on a fuel-cycle-specific basis.

In Table 2-1, locations of the lower shell B (bottom head ring) to lower vessel head circumferential weld, lower shell A to lower shell B circumferential weld, intermediate shell to lower shell A circumferential weld, and upper shell to intermediate shell circumferential weld, and outlet/inlet nozzle to upper shell welds are given for Catawba Unit 1. These locations are given relative to the origin of the radiation transport model. Note that the Catawba Unit 1 RV does not have any longitudinal welds.

Selected results from the neutron transport analyses are provided in Tables 2-2 and 2-3. In Table 2-2, calculated fast neutron ($E > 1.0$ MeV) fluence for selected RV materials at the pressure vessel clad/base metal interface is provided at future projections to 26, 34, 48 and 54 EFPY. Cycle-specific calculations were performed for Cycles 1 to 21, where a core thermal power of 3411 MWt was used. The projections were based on the assumption that the core power distributions and associated plant operating characteristics from Cycle 22 will be representative of future plant operation with an uprated core power at 3469 MWt. In Table 2-3, calculated fast neutron ($E > 1.0$ MeV) fluence at the pressure vessel clad/base metal interface is provided for Cycles 1 through 21 and future projections for Catawba Unit 1, at various azimuthal locations. Please note that the fluence values reported in Tables 2-2 and 2-3 did not consider 0.11 EFPY of neutron exposure resulting from pre-commercial operation of Catawba Unit 1, between January 1985 and June 1985. The impact of this omission on the fluence evaluation results has been assessed to be negligible (less than 0.3% of the cumulative fast neutron fluence at 60 EFPY). Future fluence evaluations will consider the pre-commercial operation phase of Catawba Unit 1.

2.3 CALCULATIONAL UNCERTAINTIES

The uncertainty associated with the calculated neutron exposure of the Catawba Unit 1 reactor pressure vessel materials is based on the recommended approach provided in Regulatory Guide 1.190. In particular, the qualification of the methodology was carried out in the following four stages:

1. Comparison of calculations with benchmark measurements from the Pool Critical Assembly (PCA) simulator at the Oak Ridge National Laboratory (ORNL).
2. Comparisons of calculations with surveillance capsule and reactor cavity measurements from the H. B. Robinson power reactor benchmark experiment.
3. An analytical sensitivity study addressing the uncertainty components resulting from important input parameters applicable to the plant-specific transport calculations used in the neutron exposure assessments.
4. Comparisons of the plant-specific calculations with all available dosimetry results from the Catawba Unit 1 surveillance program.

The first phase of the methods qualification (PCA comparisons) addressed the adequacy of basic transport calculation and dosimetry evaluation techniques and associated cross sections. This phase, however, did not test the accuracy of commercial core neutron source calculations nor did it address uncertainties in operational or geometric variables that impact power reactor calculations. The second phase of the qualification (H. B. Robinson comparisons) addressed uncertainties in these additional areas that are primarily method-related and would tend to apply generically to all fast neutron exposure evaluations.

The third phase of the qualification (analytical sensitivity study) identified the potential uncertainties introduced into the overall evaluation due to calculational methods approximations as well as to a lack of knowledge relative to various plant-specific input parameters. The overall calculational uncertainty applicable to the Catawba Unit 1 analyses was established from results of these three phases of the methods qualification.

The fourth phase of the uncertainty assessment (comparisons with Catawba Unit 1 measurements) was used solely to demonstrate the validity of the transport calculations and to confirm the uncertainty estimates associated with the analytical results. The comparison was used only as a check and was not used in any way to modify the calculated surveillance capsule and pressure vessel neutron exposures.

Table 2-4 summarizes the uncertainties developed from the first three phases of the methodology qualification. Additional information pertinent to these evaluations is provided in Reference 6. The net calculational uncertainty was determined by combining the individual components in quadrature. Therefore, the resultant uncertainty was treated as random and no systematic bias was applied to the analytical results. The plant-specific measurement comparisons given in Appendix C of this report support these uncertainty assessments for Catawba Unit 1.

Table 2-1 Pressure Vessel Material Locations for Catawba Unit 1			
Material	Axial Location ^(b) (cm)	Azimuthal Location (°)	r-θ Neutron Pad Configuration used in Exposure Calculations
Lower shell B (bottom head ring) to lower vessel head circumferential weld	-312.577	0 to 360	12.5° neutron pad
Lower shell A to lower shell B (bottom head ring) circumferential weld	-208.877	0 to 360	12.5° neutron pad
Intermediate shell to lower shell A circumferential weld	12.023	0 to 360	12.5° neutron pad
Upper shell to intermediate shell circumferential weld	224.723	0 to 360	12.5° neutron pad
Outlet Nozzle to Upper Shell Weld 1 ^(a)	275.713	22	12.5° neutron pad
Outlet Nozzle to Upper Shell Weld 2 ^(a)		158	12.5° neutron pad
Outlet Nozzle to Upper Shell Weld 3 ^(a)		202	12.5° neutron pad
Outlet Nozzle to Upper Shell Weld 4 ^(a)		338	12.5° neutron pad
Inlet Nozzle to Upper Shell Weld 1 ^(a)	264.713	67	12.5° neutron pad
Inlet Nozzle to Upper Shell Weld 2 ^(a)		113	12.5° neutron pad
Inlet Nozzle to Upper Shell Weld 3 ^(a)		247	12.5° neutron pad
Inlet Nozzle to Upper Shell Weld 4 ^(a)		293	12.5° neutron pad
Notes:			
(a) Lowest extent			
(b) Relative to the origin of the radiation transport model			

Table 2-2 Catawba Unit 1 Calculated Neutron Fluence Projections at the RV Clad/Base Metal Interface at 26, 34, 48, and 54 EFPY

RV Material	Fluence ^(a) (n/cm ² , E > 1.0 MeV)			
	26 EFPY	34 EFPY	48 EFPY	54 EFPY
Outlet Nozzle to Upper Shell Welds (Lowest Extent) 1, 2, 3, and 4	1.49E+16	1.85E+16	2.47E+16	2.74E+16
Inlet Nozzle to Upper Shell Welds (Lowest Extent) 1, 2, 3, and 4	3.08E+16	3.82E+16	5.12E+16	5.67E+16
Upper Shell Forging	6.57E+17	8.01E+17	1.05E+18	1.16E+18
Intermediate Shell Forging	1.37E+19	1.72E+19	2.34E+19	2.60E+19
Lower Shell Forging A	1.37E+19	1.72E+19	2.34E+19	2.60E+19
Lower Shell Forging B (Bottom Head Ring)	1.10E+18	1.34E+18	1.77E+18	1.95E+18
Upper Shell to Intermediate Shell Circumferential Weld	6.57E+17	8.01E+17	1.05E+18	1.16E+18
Intermediate Shell to Lower Shell A Circumferential Weld	1.37E+19	1.72E+19	2.34E+19	2.60E+19
Lower Shell A to Lower Shell B Circumferential Weld	1.10E+18	1.34E+18	1.77E+18	1.95E+18
Lower Shell B to Lower Vessel Head Circumferential Weld	1.65E+14	2.03E+14	2.71E+14	3.00E+14
Lower Shell B to Lower Vessel Head Circumferential Weld at Outside of Reactor Pressure Vessel ^(b)	2.99E+15	3.73E+15	5.02E+15	5.58E+15

Notes:

- (a) Extended beltline materials are currently interpreted to be the RV materials that will be exposed to a neutron fluence greater than or equal to 1×10^{17} n/cm² (E > 1.0 MeV) at the end of design life of the vessel (54 EFPY). Only the materials that are projected to experience a fluence value of at least 1×10^{17} n/cm² (E > 1.0 MeV) will be included in the subsequent evaluations contained within this report.
- (b) For locations far away from the reactor active core midplane, the neutron streaming along the reactor cavity causes the maximum fast neutron fluence exposure occur at outside of the reactor pressure vessel instead of clad/base metal interface.

Table 2-3 Catawba Unit 1 Calculated Neutron Fluence at the RV Clad/Base Metal Interface for Cycles 1 through 22 and Future Projections

Cycle ID	Cumulative Cycle Time (EFPY)	Fluence (n/cm ² , E > 1.0 MeV)						Maximum Fluence (n/cm ² , E > 1.0 MeV)
		0°	15°	21°	22°	30°	45°	
1	0.79	3.67E+17	5.64E+17	6.71E+17	6.74E+17	6.66E+17	7.59E+17	7.59E+17
2	1.54	7.19E+17	1.02E+18	1.14E+18	1.14E+18	1.11E+18	1.18E+18	1.18E+18
3	2.31	1.04E+18	1.51E+18	1.70E+18	1.70E+18	1.61E+18	1.67E+18	1.70E+18
4	3.17	1.42E+18	2.03E+18	2.28E+18	2.28E+18	2.13E+18	2.15E+18	2.28E+18
5	3.96	1.74E+18	2.49E+18	2.79E+18	2.78E+18	2.60E+18	2.62E+18	2.79E+18
6	4.98	2.17E+18	3.13E+18	3.52E+18	3.51E+18	3.21E+18	3.15E+18	3.52E+18
7	5.93	2.53E+18	3.68E+18	4.15E+18	4.13E+18	3.79E+18	3.73E+18	4.15E+18
8	7.00	2.90E+18	4.30E+18	4.87E+18	4.85E+18	4.41E+18	4.24E+18	4.87E+18
9	8.17	3.28E+18	4.93E+18	5.59E+18	5.56E+18	5.03E+18	4.80E+18	5.59E+18
10	9.29	3.63E+18	5.47E+18	6.21E+18	6.19E+18	5.66E+18	5.45E+18	6.21E+18
11	10.48	3.94E+18	5.94E+18	6.76E+18	6.74E+18	6.19E+18	5.95E+18	6.76E+18
12	11.85	4.32E+18	6.52E+18	7.45E+18	7.43E+18	6.91E+18	6.74E+18	7.45E+18
13	13.25	4.70E+18	7.10E+18	8.13E+18	8.11E+18	7.59E+18	7.50E+18	8.13E+18
14	14.69	5.06E+18	7.64E+18	8.77E+18	8.76E+18	8.26E+18	8.22E+18	8.77E+18
15	15.99	5.40E+18	8.18E+18	9.41E+18	9.40E+18	8.86E+18	8.74E+18	9.41E+18
16	17.35	5.71E+18	8.70E+18	1.00E+19	1.00E+19	9.50E+18	9.36E+18	1.00E+19
17	18.68	6.02E+18	9.18E+18	1.06E+19	1.06E+19	1.01E+19	9.90E+18	1.06E+19
18	20.05	6.36E+18	9.69E+18	1.12E+19	1.12E+19	1.06E+19	1.04E+19	1.12E+19
19	21.38	6.68E+18	1.02E+19	1.18E+19	1.18E+19	1.12E+19	1.09E+19	1.18E+19
20	22.82	7.01E+18	1.07E+19	1.24E+19	1.24E+19	1.17E+19	1.14E+19	1.24E+19
21	24.18	7.33E+18	1.11E+19	1.29E+19	1.29E+19	1.23E+19	1.21E+19	1.29E+19
22	25.62	7.67E+18	1.17E+19	1.35E+19	1.35E+19	1.29E+19	1.26E+19	1.35E+19
---	26.00	7.76E+18	1.18E+19	1.37E+19	1.37E+19	1.30E+19	1.27E+19	1.37E+19
---	30.00	8.73E+18	1.33E+19	1.54E+19	1.55E+19	1.47E+19	1.42E+19	1.55E+19
---	34.00	9.69E+18	1.48E+19	1.72E+19	1.72E+19	1.63E+19	1.56E+19	1.72E+19
---	40.00	1.11E+19	1.70E+19	1.98E+19	1.99E+19	1.88E+19	1.78E+19	1.99E+19
---	44.00	1.21E+19	1.85E+19	2.16E+19	2.16E+19	2.05E+19	1.93E+19	2.16E+19
---	48.00	1.31E+19	2.00E+19	2.34E+19	2.34E+19	2.21E+19	2.07E+19	2.34E+19
---	54.00	1.45E+19	2.22E+19	2.60E+19	2.60E+19	2.46E+19	2.29E+19	2.60E+19
---	60.00	1.60E+19	2.45E+19	2.86E+19	2.87E+19	2.71E+19	2.51E+19	2.87E+19

Table 2-4 Calculational Uncertainties		
Description	Uncertainty	
	Capsule	Vessel IR
PCA Comparisons	3%	3%
H. B. Robinson Comparisons	3%	3%
Analytical Sensitivity Studies	10%	11%
Additional Uncertainty for Factors not Explicitly Evaluated	5%	5%
Net Calculational Uncertainty	12%	13%

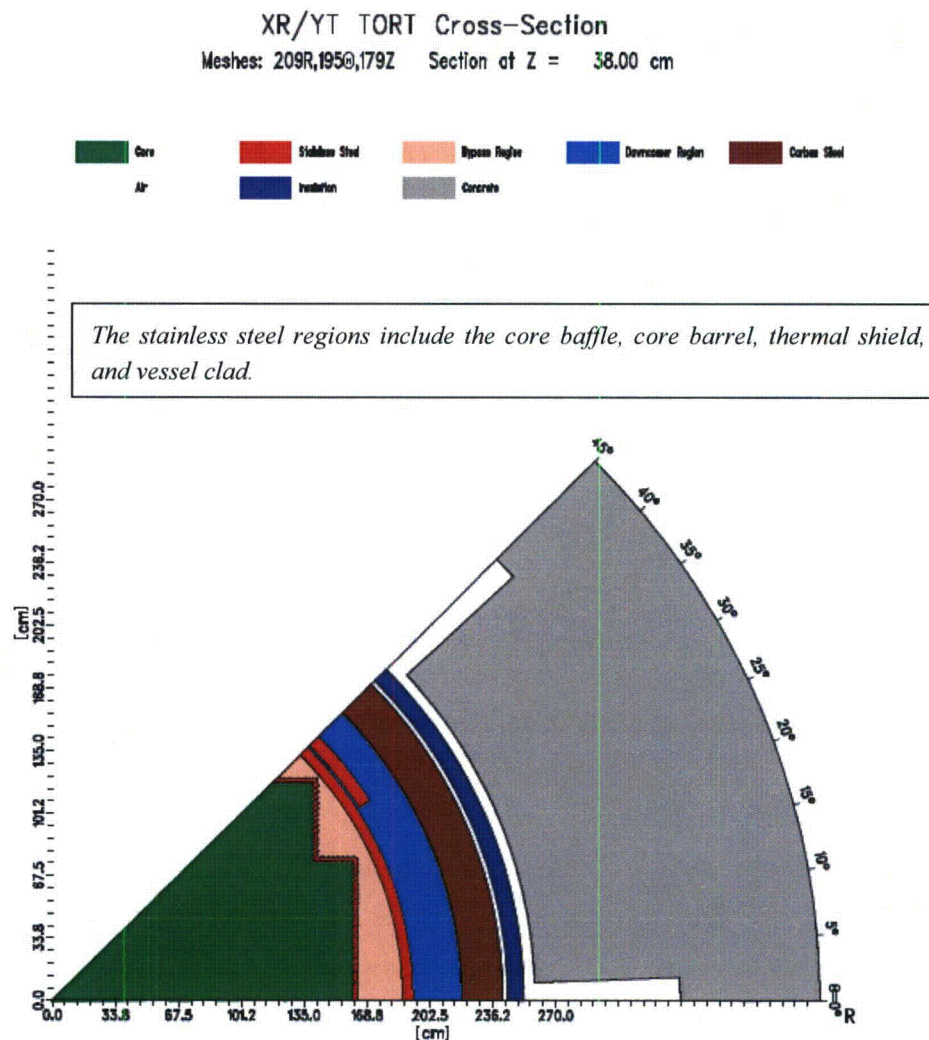


Figure 2-1 Catawba Unit 1 Reactor Geometry in r-θ at the Core Mid-plane – 12.5° Neutron Pad Configuration

XR/YT TORT Cross-Section
 Meshes: 209R,195Θ,179Z Section at Z = 38.00 cm



The stainless steel regions include the core baffle, core barrel, neutron pad, surveillance capsule holder, and vessel clad. The carbon steel regions include the surveillance capsule specimens and pressure vessel.

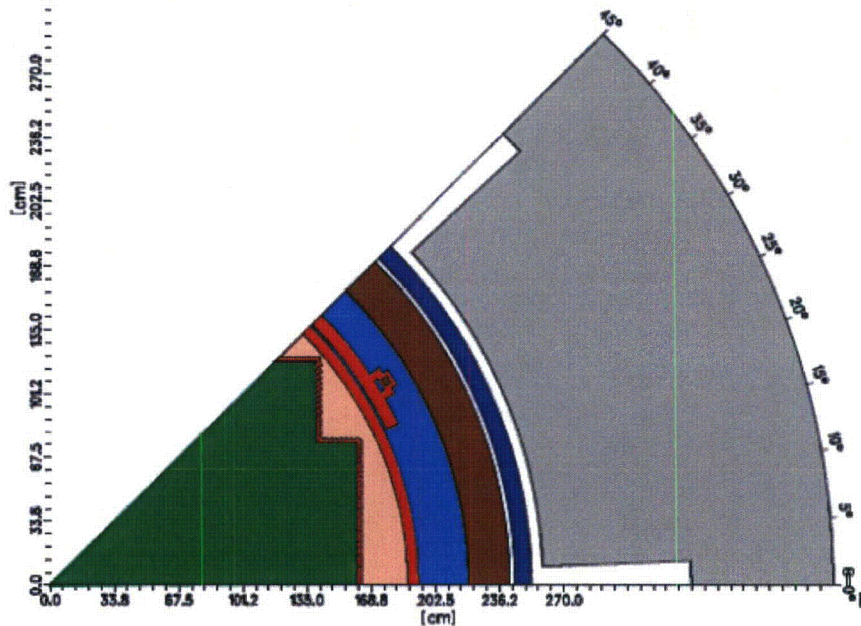


Figure 2-2 Catawba Unit 1 Reactor Geometry in r-θ at the Core Mid-plane – 20.0° Neutron Pad Configuration

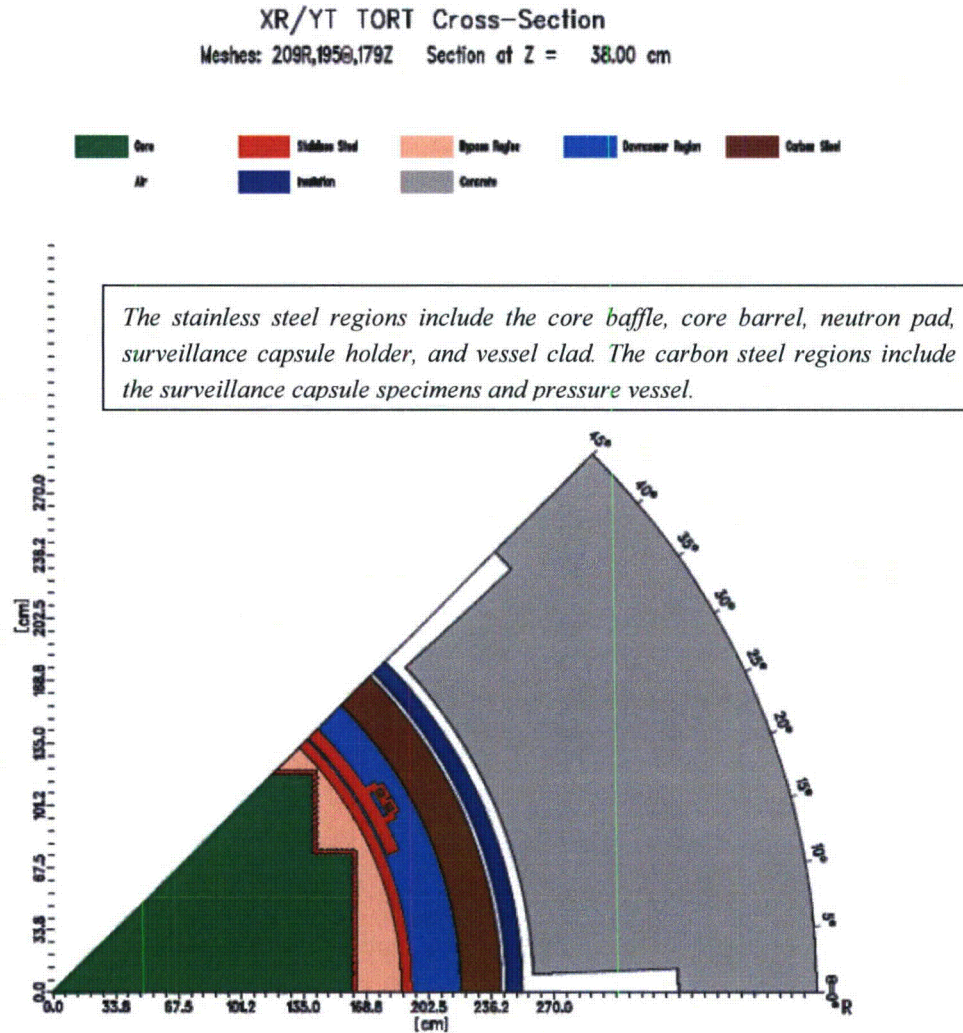


Figure 2-3 Catawba Unit 1 Reactor Geometry in r-θ at the Core Mid-plane – 22.5° Neutron Pad Configuration

R/Z TORT Cross-Section
Meshes: 209R,195 θ ,179Z Section at angle $\theta = 40^\circ$

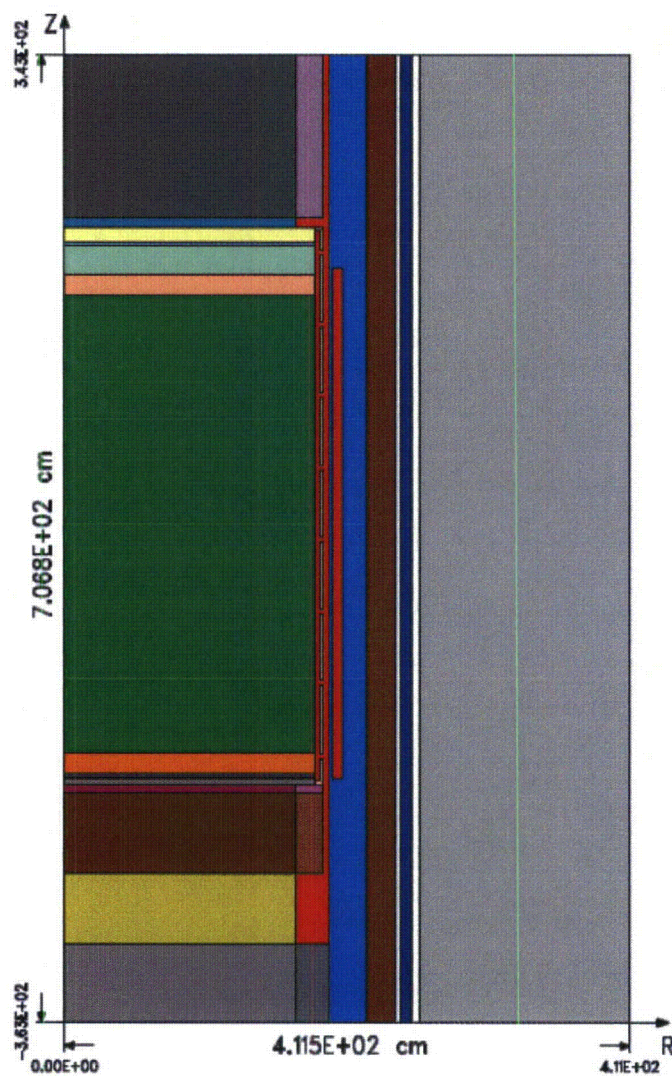


Figure 2-4 Catawba Unit 1 Reactor Geometry in r-z Plane at 40° Azimuthal Angle

R/Z TORT Cross-Section
Meshes: 209R,195B,179Z Section at angle $\theta = 29^\circ$

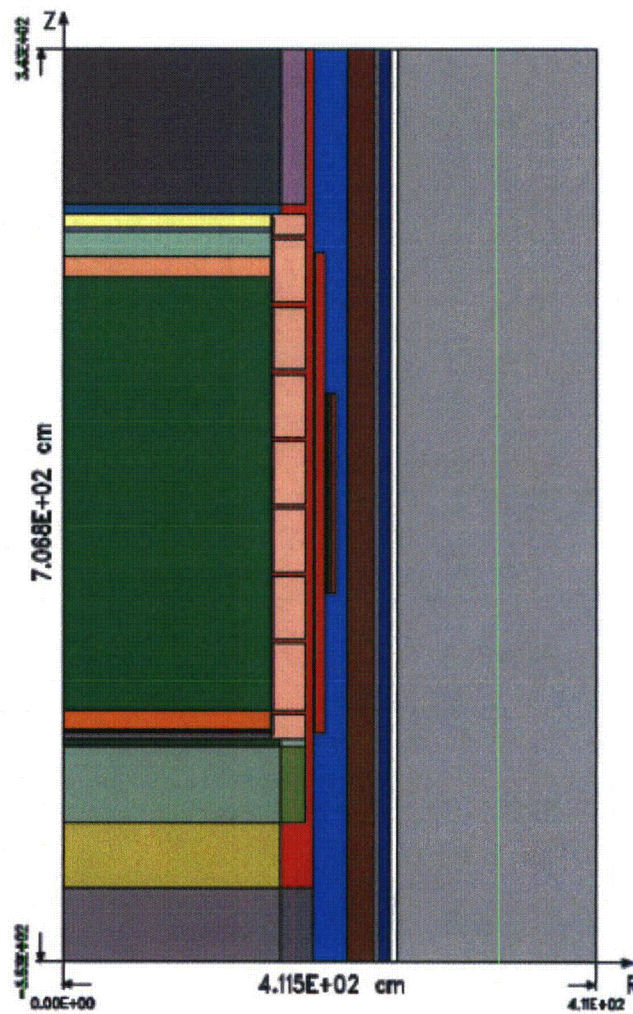


Figure 2-5 Catawba Unit 1 Reactor Geometry in r-z Plane at 29° Azimuthal Angle

3 MATERIAL PROPERTY INPUT

The fracture toughness properties of the ferritic materials in the reactor coolant pressure boundary are determined in accordance with the NRC Standard Review Plan (Reference 10). The beltline region of a RV, per 10 CFR 50.61 (Reference 3), is defined as:

“the region of the reactor vessel (shell material including welds, heat affected zones and plates or forgings) that directly surrounds the effective height of the active core and adjacent regions of the reactor vessel that are predicted to experience sufficient neutron radiation damage to be considered in the selection of the most limiting material with regard to radiation damage.”

The beltline materials, as described in the paragraph above, must be considered in the RV integrity evaluations. Additionally, as described in Item IV.A2.R-84 of NUREG-1801, Revision 2 (Reference 11), any materials with an EOLE fluence value exceeding $1.0 \times 10^{17} \text{ n/cm}^2$ ($E > 1.0 \text{ MeV}$) must be considered in the RV integrity evaluations. The materials that exceed this threshold are referred to as extended beltline materials and are evaluated to ensure that the applicable acceptance criteria are met through EOLE.

Summaries of the best-estimate (BE) copper and nickel contents, $RT_{NDT(U)}$ values, and initial USE values for the RV materials are provided in Table 3-1 for Catawba Unit 1.

Table 3-1 Material Properties for the Catawba Unit 1 RV^(a)				
Material Description	Chemical Composition		Fracture Toughness Properties	
RV Material and Identification Number	Cu Wt. %	Ni Wt. %	Initial RT_{NDT}^(b) (°F)	Initial USE (ft-lb)
Upper Shell (US) Forging 06	0.16 ^(c)	0.85	-26	101
Intermediate Shell (IS) Forging 05	0.09	0.86	-8	134
Lower Shell (LS) Forging 04	0.04 ^(d)	0.83	-13	134
Bottom Head Ring 03	0.06	0.77	14	68
US to IS Circumferential (Circ.) Weld W06 (Heat # 899680)	0.03 ^(e)	0.75 ^(e)	10 ^(e)	92 ^(f)
IS to LS Circ. Weld W05 (Heat # 895075)	0.04	0.72	-51	130
LS to Bottom Head Ring Weld W04 (Heat # 899680)	0.03 ^(e)	0.75 ^(e)	10 ^(e)	92 ^(f)
Catawba Unit 1 Surveillance Weld (Heat # 895075)	0.05 ^(g)	0.73 ^(g)	---	---
McGuire Unit 2 Surveillance Weld (Heat # 895075)	0.04 ^(g)	0.74 ^(g)	---	---
Watts Bar Unit 1 Surveillance Weld (Heat # 895075)	0.03 ^(g)	0.75 ^(g)	---	---
Notes: (a) Values obtained from Tables 3.1.2-1 and 3.1.2-2 of WCAP-17175-P (Reference 12), unless otherwise noted. (b) All initial RT _{NDT} values are based on measured data. (c) No weight percent copper value was reported in the Certified Material Test Report (CMTR). Therefore, the maximum copper weight percent value for A508 Class 2 forging materials is conservatively applied based on the generic data provided in Appendix G of ORNL document ORNL/TM-2006/530 (Reference 14). (d) According to WCAP-17175-P (Reference 12), the weight percent copper value is 0.05. However, according to the CMTR, the weight percent copper value is 0.04, which is consistent with the value used in the Catawba License Renewal Application (LRA, [Reference 13]). Therefore, 0.04 wt. % copper is utilized in this evaluation. (e) Values are based on measured data from the weld certification records for weld Heat # 899680. The initial RT _{NDT} was determined using the measured data and the method described in Section B.1.1(4) of NUREG-0800 Branch Technical Position 5-3 (Reference 10). Note that the RT _{NDT} for this value is higher than that reported in Table 3.1.2-2 of Reference 12; however, based on the CMTR data, the Charpy tests were performed at 10°F; therefore, the initial RT _{NDT} should be 10°F, rather than 0°F. (f) Value obtained from Table 4.2-3 of the McGuire and Catawba LRA. (g) Information for the surveillance welds is taken from Table 4 of WCAP-15203, Revision 1 (Reference 15).				

4 SURVEILLANCE DATA

Per 10 CFR 50.61, calculation of Position 2.1 chemistry factors requires data from the plant-specific surveillance program. Furthermore, Regulatory Guide 1.99, Revision 2 allows the use of data from the plant-specific surveillance program in determining the decrease in USE.

In addition to the plant-specific surveillance data, 10 CFR 50.61 also requires the use of data from surveillance programs at other plants which include the same limiting beltline material when calculating Position 2.1 chemistry factors. Data from a surveillance program at another plant are often called 'sister plant' data.

Tables 4-1 and 4-2 summarize the Catawba Unit 1 surveillance data as well as surveillance data from McGuire Unit 2 and Watts Bar Unit 1. The McGuire Unit 2 and Watts Bar Unit 1 surveillance programs include weld Heat # 895075, which is the same weld heat as the Catawba Unit 1 Intermediate Shell to Lower Shell circumferential weld. Thus, the McGuire Unit 2 and Watts Bar Unit 1 data will be used in calculation of the Position 2.1 chemistry factor value for Catawba Unit 1 weld Heat # 895075.

Table 4-1 Catawba Unit 1 Surveillance Capsule Data

Material	Capsule	Withdrawal EFPY	Lead Factor ^(a)	Capsule Fluence ^(a) ($\times 10^{19}$ n/cm ² , E > 1.0 MeV)	Measured 30 ft-lb Transition Temperature Shift ^(b) (°F)	Measured USE Decrease ^(b) (%)
IS Forging 05 (Axial)	Z	0.79	3.85	0.292	15.74	0 ^(d)
	Y	4.98	3.73	1.31	48.63	4
	V	9.29	3.72	2.31	50.58	1
	X ^(c)	9.29	3.88	2.41	---	---
	U ^(c)	9.29	3.88	2.41	---	---
	W ^(c)	14.69	4.00	3.51	---	---
IS Forging 05 (Tangential)	Z	0.79	3.85	0.292	0.0 ^(e)	0
	Y	4.98	3.73	1.31	19.09	9
	V	9.29	3.72	2.31	25.61	10
	X ^(c)	9.29	3.88	2.41	---	---
	U ^(c)	9.29	3.88	2.41	---	---
	W ^(c)	14.69	4.00	3.51	---	---
Surveillance Weld Material (Heat # 895075)	Z	0.79	3.85	0.292	1.91	5
	Y	4.98	3.73	1.31	17.79	2
	V	9.29	3.72	2.31	26.5	3
	X ^(c)	9.29	3.88	2.41	---	---
	U ^(c)	9.29	3.88	2.41	---	---
	W ^(c)	14.69	4.00	3.51	---	---

Notes:

- (a) The calculated fluence values and lead factors were updated as part of the MUR power uprate analysis.
- (b) Values obtained from Table 5-10 of WCAP-15117 (Reference 16).
- (c) Capsules X and U were removed and the dosimeters were analyzed, but the specimens were not. Capsule W was placed in the spent fuel pool following removal.
- (d) Original value was measured as a +1 % increase, but an increase should not occur; therefore, a conservative value of zero will be used.
- (e) Original value was -14.9°F, but physically a reduction should not occur; therefore, a value of zero will be used.

Table 4-2 McGuire Unit 2 and Watts Bar Unit 1 Surveillance Capsule Data for Weld Heat # 895075

Material	Capsule	Capsule Fluence ($\times 10^{19}$ n/cm ² , E > 1.0 MeV)	Measured 30 ft-lb Transition Temperature Shift (°F)	Inlet Temperature (°F)	Temperature Adjustment (°F)
McGuire Unit 2 Surveillance Weld ^(a) (Heat # 895075)	V	0.302	38.51	557 ^(b)	+4 ^(d)
	X	1.38	35.93		
	U	1.90	23.81		
	W	2.82	43.76		
Watts Bar Unit 1 Surveillance Weld ^(a) (Heat # 895075)	U	0.447	0.0 ^(c)	560	+7 ^(d)
	W	1.08	30.5		
	X	1.71	25.8		
	Z	2.40	13.9		

Notes:

- (a) Data pertaining to the McGuire Unit 2 and Watts Bar Unit 1 surveillance welds were taken from Tables 4.2-1 and 4.2-2 of WCAP-17455-NP (Reference 17), respectively, unless otherwise noted.
- (b) McGuire Unit 2 capsules were irradiated between Cycles 1 through 10. An average inlet temperature of 557°F was determined over the period of capsule irradiation.
- (c) Original value was -6.4°F, but physically a reduction should not occur; therefore, a value of zero will be used.
- (d) Temperature adjustment = $1.0 \times (T_{\text{capsule}} - T_{\text{plant}})$, where $T_{\text{plant}} = 553^\circ\text{F}$ for Catawba Unit 1 (applied to the weld ΔRT_{NDT} data for each of the McGuire Unit 2 and Watts Bar Unit 1 capsules in the Position 2.1 Chemistry Factor calculation).

5 CHEMISTRY FACTORS

As described in Section 1 of this report, Position 1.1 chemistry factors for each RV material are calculated using the best-estimate copper and nickel weight percent of the material and Tables 1 and 2 of 10 CFR 50.61. The best-estimate copper and nickel weight percents for the Catawba Unit 1 RV materials were provided in Table 3-1 of this report.

The Position 2.1 chemistry factors are calculated for the materials that have available surveillance program results. The calculation is performed using the method described in 10 CFR 50.61, which is also summarized in Section 1 of this report. The Catawba Unit 1 surveillance data as well as any applicable sister plant data were summarized in Section 4 of this report, and will be utilized in the Position 2.1 chemistry factor calculations in this Section.

The Position 2.1 chemistry factor calculations are presented in Table 5-1 for the Catawba Unit 1 RV materials contained in the radiation surveillance program. These values were calculated using the surveillance data summarized in Section 4 of this report. Additionally, surveillance data from McGuire Unit 2 and Watts Bar Unit 1 are utilized in Table 5-1 as it is applicable to the Catawba Unit 1 Intermediate Shell to Lower Shell circumferential weld (Heat # 895075).

All of the surveillance weld data are adjusted for irradiation temperature and chemical composition differences in accordance with the guidance presented at an industry meeting held by the NRC on February 12 and 13, 1998 (Reference 18).

The Position 1.1 chemistry factors are summarized along with the Position 2.1 chemistry factors in Table 5-2 for Catawba Unit 1.

Appendix A contains the credibility evaluation for each of the surveillance materials for which a chemistry factor is calculated in this Section. Margin will be applied to the calculations of ART and RT_{PTS} according to the conclusions of the credibility evaluation for each of the surveillance materials.

Table 5-1 Calculation of Catawba Unit 1 Position 2.1 Chemistry Factor Values Using Surveillance Capsule Test Results

Material	Capsule	Capsule $f^{(a)}$ ($\times 10^{19}$ n/cm ² , E > 1.0 MeV)	FF ^(b)	$\Delta RT_{NDT}^{(c)}$ (°F)	FF * ΔRT_{NDT} (°F)	FF ²
IS Forging 05 (Axial)	Z	0.292	0.663	15.74	10.44	0.440
	Y	1.31	1.075	48.63	52.28	1.156
	V	2.31	1.226	50.58	62.03	1.504
IS Forging 05 (Tangential)	Z	0.292	0.663	0.0 ^(d)	0.00	0.440
	Y	1.31	1.075	19.09	20.52	1.156
	V	2.31	1.226	25.61	31.41	1.504
SUM:					176.68	6.199
$CF_{IS\ Forging} = \Sigma(FF * \Delta RT_{NDT}) \div \Sigma(FF^2) = (176.68) \div (6.199) = 28.5^{\circ}F$						
Catawba Unit 1 Surveillance Weld (Heat # 895075)	Z	0.292	0.663	1.51 (1.91)	1.00	0.440
	Y	1.31	1.075	14.05 (17.79)	15.11	1.156
	V	2.31	1.226	20.94 (26.5)	25.67	1.504
McGuire Unit 2 Surveillance Weld (Heat # 895075)	V	0.302	0.672	42.51 (38.51)	28.57	0.452
	X	1.38	1.089	39.93 (35.93)	43.50	1.187
	U	1.90	1.176	27.81 (23.81)	32.70	1.382
	W	2.82	1.276	47.76 (43.76)	60.93	1.628
Watts Bar Unit 1 Surveillance Weld (Heat # 895075)	U	0.447	0.776	9.24 (0.0 ^(d))	7.17	0.602
	W	1.08	1.022	49.50 (30.5)	50.57	1.044
	X	1.71	1.148	43.30 (25.8)	49.69	1.317
	Z	2.40	1.236	27.59 (13.9)	34.10	1.528
SUM:					349.00	12.238
$CF_{Weld\ Metal} = \Sigma(FF * \Delta RT_{NDT}) \div \Sigma(FF^2) = (349.00) \div (12.238) = 28.5^{\circ}F$						
Notes: (a) f = fluence. (b) FF = fluence factor = $f^{(0.28 - 0.10 \cdot \log f)}$. (c) ΔRT_{NDT} values are the measured 30 ft-lb shift values. All values are taken from Tables 4-1 and 4-2 of this report. The McGuire Unit 2 and Watts Bar Unit 1 surveillance weld ΔRT_{NDT} values have been adjusted according to the temperature adjustments summarized in Table 4-2 of this report then by using the ratio procedure to account for differences in the surveillance weld chemistry and the beltline weld chemistry (pre-adjusted values are listed in parentheses). For McGuire Unit 2, Ratio = $CF_{Vessel\ Weld} / CF_{Surv.\ Weld} = 54^{\circ}F / 54^{\circ}F = 1.00$. For Watts Bar Unit 1, Ratio = $54^{\circ}F / 41^{\circ}F = 1.32$. The Catawba Unit 1 surveillance weld ΔRT_{NDT} values are not adjusted for temperature differences, but are adjusted by a ratio for chemistry differences (pre-adjusted values are listed in parentheses). For Catawba Unit 1, Ratio = $54^{\circ}F / 68^{\circ}F = 0.79$. (d) This ΔRT_{NDT} value was determined to be negative, but physically a reduction should not occur; therefore, a value of zero is used.						

Table 5-2 Summary of Catawba Unit 1 Positions 1.1 and 2.1 Chemistry Factors

RV Material and Identification Number	Chemistry Factor (°F)	
	Position 1.1 ^(a)	Position 2.1 ^(b)
US Forging 06	123.5	---
IS Forging 05	58	28.5
LS Forging 04	26	---
Bottom Head Ring 03	37	---
US to IS Circ. Weld W06 (Heat # 899680)	41	---
IS to LS Circ. Weld W05 (Heat # 895075)	54	28.5
LS to Bottom Head Ring Weld W04 (Heat # 899680)	41	---
Catawba Unit 1 Surveillance Weld (Heat # 895075)	68	---
McGuire Unit 2 Surveillance Weld (Heat # 895075)	54	---
Watts Bar Unit 1 Surveillance Weld (Heat # 895075)	41	---
Notes: (a) Position 1.1 Chemistry Factors were calculated using the copper and nickel weight percents presented in Table 3-1 of this report and Tables 1 and 2 of 10 CFR 50.61. (b) Position 2.1 Chemistry Factors taken from Table 5-1 of this report. Per Appendix A, the Catawba Unit 1 surveillance forging and weld data are credible.		

6 PRESSURIZED THERMAL SHOCK CALCULATIONS

A limiting condition on RV integrity known as PTS may occur during a severe system transient such as a loss-of-coolant accident (LOCA) or steam line break. Such transients may challenge the integrity of the RV under the following conditions:

- Severe overcooling of the inside surface of the vessel wall followed by high repressurization,
- Significant degradation of vessel material toughness caused by radiation embrittlement,
- The presence of a critical-size defect anywhere within the vessel wall.

In 1985, the U.S. NRC issued a formal ruling (10 CFR 50.61) on PTS (Reference 3) that established screening criteria on RV embrittlement, as measured by the maximum RT_{NDT} in the limiting beltline component at the end-of-license, termed reference temperature for PTS (RT_{PTS}). RT_{PTS} screening values were set by the U.S. NRC for beltline axial welds, forgings or plates, and for beltline circumferential weld seams for plant operation to the end of plant license. All domestic pressurized water reactor (PWR) vessels have been required to evaluate vessel embrittlement in accordance with the criteria through the end-of-license. The U.S. NRC revised 10 CFR 50.61 in 1991 and 1995 to change the procedure for calculating radiation embrittlement. These revisions made the procedure for calculating the RT_{PTS} values consistent with the methods given in Regulatory Guide 1.99, Revision 2 (Reference 1).

These accepted methods were used with the surface fluence of Section 2 to calculate the following RT_{PTS} values for the Catawba Unit 1 RV materials at 54 EFPY (EOLE). The EOLE RT_{PTS} calculations are summarized below in Table 6-1 for Catawba Unit 1.

PTS Conclusion

For Catawba Unit 1, the limiting RT_{PTS} value at 54 EFPY is 63°F (see Table 6-1); this value corresponds to US Forging 06. Therefore, all of the beltline and extended beltline materials in the Catawba Unit 1 reactor vessel are below the RT_{PTS} screening criteria values of 270°F for forgings, and 300°F for circumferentially oriented welds through EOLE (54 EFPY).

The Alternate PTS Rule (10 CFR 50.61a [Reference 19]) was published in the *Federal Register* by the NRC in 2010. This alternate rule is less restrictive than the Mandatory PTS Rule (10 CFR 50.61) and is intended to be used for situations where the 10 CFR 50.61 criteria cannot be met. Catawba Unit 1 currently meets the criteria for the Mandatory PTS Rule through EOLE and therefore does not need to utilize the Alternate PTS Rule at this time.

Table 6-1 RT_{PTS} Calculations for the Catawba Unit 1 RV Materials at 54 EFPY^(a)

RV Material and Identification Number	R.G. 1.99, Rev. 2 Position	CF ^(b) (°F)	Fluence ^(c) ($\times 10^{19}$ n/cm ² , E > 1.0 MeV)	FF	IRT _{NDT} ^(d) (°F)	Δ RT _{NDT} (°F)	σ_U ^(d) (°F)	σ_Δ ^(e) (°F)	Margin (°F)	RT _{PTS} (°F)
US Forging 06	1.1	123.5	0.116	0.4472	-26	55.2	0	17.0	34.0	63
IS Forging 05	1.1	58	2.60	1.2559	-8	72.8	0	17.0	34.0	99
<i>Using credible surveillance data</i>	2.1	28.5	2.60	1.2559	-8	35.8	0	8.5	17.0	45
LS Forging 04	1.1	26	2.60	1.2559	-13	32.7	0	16.3	32.7	52
Bottom Head Ring 03	1.1	37	0.195	0.5634	14	20.8	0	10.4	20.8	56
US to IS Circ. Weld W06 (Heat # 899680)	1.1	41	0.116	0.4472	10	18.3	0	9.2	18.3	47
IS to LS Circ. Weld W05 (Heat # 895075)	1.1	54	2.60	1.2559	-51	67.8	0	28.0	56.0	73
<i>Using credible surveillance data</i>	2.1	28.5	2.60	1.2559	-51	35.8	0	14.0	28.0	13
LS to Bottom Head Ring Weld W04 (Heat # 899680)	1.1	41	0.195	0.5634	10	23.1	0	11.5	23.1	56

Notes:

- (a) The 10 CFR 50.61 methodology was utilized in the calculation of the RT_{PTS} values. See Section 1 of this report for details.
- (b) Taken from Table 5-2 of this report.
- (c) Taken from Table 2-2 of this report.
- (d) Initial RT_{NDT} values are measured and are taken from Table 3-1 of this report.
- (e) Per Appendix A of this report, the surveillance data of the forging and weld were deemed credible. Per the guidance of 10 CFR 50.61, the base metal $\sigma_\Delta = 17^\circ\text{F}$ for Position 1.1 and, with credible surveillance data, $\sigma_\Delta = 8.5^\circ\text{F}$ for Position 2.1; the weld metal $\sigma_\Delta = 28^\circ\text{F}$ for Position 1.1 and, with credible surveillance data, $\sigma_\Delta = 14^\circ\text{F}$ for Position 2.1. However, σ_Δ need not exceed $0.5 \cdot \Delta\text{RT}_{\text{NDT}}$.

7 UPPER-SHELF ENERGY CALCULATIONS

The requirements for USE are contained in 10 CFR 50, Appendix G (Reference 20). 10 CFR 50, Appendix G requires utilities to submit an analysis at least 3 years prior to the time that the USE of any RV material is predicted to drop below 50 ft-lb.

Regulatory Guide 1.99, Revision 2 defines two methods that can be used to predict the decrease in USE due to irradiation. The method to be used depends on the availability of credible surveillance capsule data. For RV beltline materials that are not in the surveillance program or are not credible, the Charpy USE (Position 1.2) is assumed to decrease as a function of fluence and copper content, as indicated in Regulatory Guide 1.99, Revision 2 (Reference 1).

When two or more credible surveillance data sets become available from the RV, they may be used to determine the Charpy USE of the surveillance materials. The surveillance data are then used in conjunction with Figure 2 of the Regulatory Guide to predict the decrease in USE (Position 2.2) of the RV materials due to irradiation. If the EOLE USE values calculated using Position 2.2 are most limiting, then they must be used regardless of the credibility of the surveillance data.

The 54 EFPY (EOLE) Position 1.2 USE values of the RV materials can be predicted using the corresponding 1/4T fluence projection, the copper content, and Figure 2 in Regulatory Guide 1.99, Revision 2.

The predicted Position 2.2 USE values are determined for the RV materials that are contained in the surveillance program by using the reduced plant surveillance data along with the corresponding 1/4T fluence projection. The reduced plant surveillance data were obtained from Table 5-10 of WCAP-15117 (Reference 16) for Catawba Unit 1. The surveillance data were plotted on Regulatory Guide 1.99, Revision 2, Figure 2 (see Figure 7-1 of this report) using the updated surveillance capsule fluence values documented in Table 4-1 of this report for Catawba Unit 1. These data were fitted by drawing a line parallel to the existing lines as the upper bound of all the surveillance data. These reduced lines were used instead of the existing lines to determine the Position 2.2 EOLE USE values.

The projected USE values were calculated to determine if the Catawba Unit 1 RV materials remain above the 50 ft-lb limit at EOLE. These calculations are summarized in Table 7-1 for Catawba Unit 1.

USE Conclusion

For Catawba Unit 1, the limiting USE value at 54 EFPY is 60 ft-lb (see Table 7-1); this value corresponds to Bottom Head Ring 03. Therefore, all of the beltline and extended beltline materials in the Catawba Unit 1 RV are projected to remain above the USE screening criterion value of 50 ft-lb (per 10 CFR 50, Appendix G) through EOLE (54 EFPY).

Table 7-1 Catawba Unit 1 Predicted Positions 1.2 and 2.2 USE Values at 54 EFPY

RV Material and Identification Number	Wt. % Cu^(a)	1/4T EOLE Fluence^(b) ($\times 10^{19}$ n/cm², E > 1.0 MeV)	Unirradiated USE^(a) (ft-lb)	Projected USE Decrease (%)	Projected EOLE USE (ft-lb)
US Forging 06	0.16	0.070	101	14	87
IS Forging 05	0.09	1.565	134	21 ^(c)	106
<i>Using surveillance data</i>	0.09	1.565	134	10 ^(d)	121
LS Forging 04	0.04	1.565	134	21 ^(c)	106
Bottom Head Ring 03	0.06	0.117	68	12 ^(c)	60
US to IS Circ. Weld W06 (Heat # 899680)	0.03	0.070	92	10 ^(c)	83
IS to LS Circ. Weld W05 (Heat # 895075)	0.04	1.565	130	21 ^(c)	103
<i>Using surveillance data</i>	0.04	1.565	130	8 ^(d)	120
LS to Bottom Head Ring Weld W04 (Heat # 899680)	0.03	0.117	92	12 ^(c)	81

Notes:

- (a) From Table 3-1 of this report.
- (b) 1/4T fluence was calculated using Equation (3) of Regulatory Guide 1.99, Revision 2, and the Catawba Unit 1 RV beltline wall thickness of 8.465 inches.
- (c) Percentage USE decrease is conservatively based on lowest Cu Wt. % chemistry line (0.05% for weld and 0.10% for base metal) delineated in Figure 2 of Regulatory Guide 1.99, Revision 2.
- (d) Percentage USE decrease is based on Position 2.2 of Regulatory Guide 1.99, Revision 2 using data from Table 4-1. Credibility Criterion 3 in the Discussion section of Regulatory Guide 1.99, Revision 2, indicates that even if the surveillance data are not considered credible for determination of ΔRT_{NDT} , "they may be credible for determining decrease in upper-shelf energy if the upper shelf can be clearly determined, following the definition given in ASTM E 185-82." Regulatory Guide 1.99, Revision 2, Position 2.2 indicates that an upper-bound line drawn parallel to the existing lines (in Figure 2 of the Guide) through the surveillance data points should be used in preference to the existing graph lines for determining the decrease in USE.

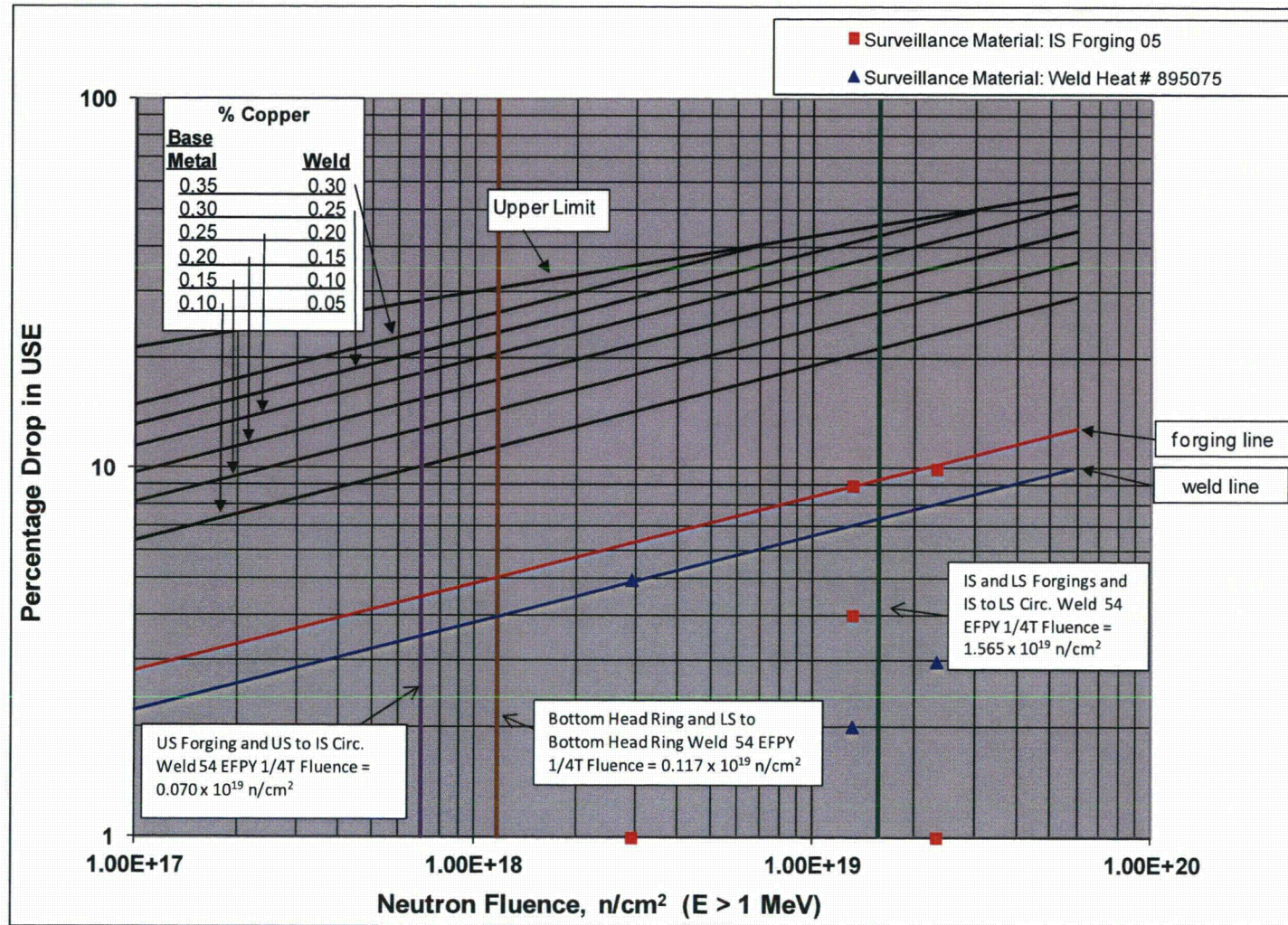


Figure 7-1 Regulatory Guide 1.99, Revision 2 Predicted Decrease in USE as a Function of Copper and Fluence for Catawba Unit 1

8 HEATUP AND COOLDOWN PRESSURE-TEMPERATURE LIMIT CURVES APPLICABILITY

Heatup and cooldown limit curves are calculated using the most limiting values of RT_{NDT} corresponding to the limiting RV material. The most limiting RV material RT_{NDT} values are determined by using the unirradiated RV material fracture toughness properties and estimating the irradiation-induced shift (ΔRT_{NDT}).

RT_{NDT} increases as the material is exposed to fast-neutron irradiation; therefore, to find the most limiting RT_{NDT} at any time period in the reactor's life, ΔRT_{NDT} due to the radiation exposure associated with that time period must be added to the original unirradiated RT_{NDT} . Using the ART values, pressure-temperature (P-T) limit curves are determined in accordance with the requirements of 10 CFR Part 50, Appendix G (Reference 20), as augmented by Appendix G to Section XI of the ASME Boiler and Pressure Vessel Code (Reference 21).

The P-T limit curves for normal heatup and cooldown of the primary reactor coolant system for Catawba Unit 1 were previously developed in WCAP-15203, Revision 1 (Reference 15) for 34 EFPY and WCAP-15448, Revision 1 (Reference 22) for 51 EFPY. The existing 34 and 51 EFPY P-T limit curves are based on the limiting beltline material ART values, which are influenced by both the fluence and the initial material properties of that material.

To determine the applicability of the Catawba Unit 1 P-T limit curves developed in WCAP-15203, Revision 1 (Reference 15) for 34 EFPY and in WCAP-15448, Revision 1 (Reference 22) for 51 EFPY, the limiting reactor vessel material ART values with consideration of the MUR power uprate are compared to the limiting beltline material ART values used in development of the existing 34 EFPY and 51 EFPY P-T limit curves contained in References 15 and 22. The Regulatory Guide 1.99, Revision 2 (Reference 1) methodology was used along with the surface fluence of Section 2 to calculate ART values for the Catawba Unit 1 reactor vessel materials at 34 EFPY and 54 EFPY. Note that the 54 EFPY ART values calculated as part of this MUR uprate evaluation will be used when assessing the applicability of the existing 51 EFPY P-T limit curves. 54 EFPY ART values (as opposed to 51 EFPY values) were calculated to correspond with the other RVI evaluations calculated as part of this MUR power uprate analysis at EOLE. The ART calculations with consideration of the MUR power uprate are summarized in Tables 8.1-1 through 8.1-4 for Catawba Unit 1.

Existing P-T Limit Curves Applicability Conclusions

Comparisons of the limiting MUR power uprate ART values to those used in calculation of the existing P-T limit curves are contained in Tables 8.2-1 and 8.2-2 for Catawba Unit 1. With a re-evaluation of surveillance data credibility, a recalculation of chemistry factors, and the consideration of all RV materials projected to achieve surface fluence levels of 1×10^{17} n/cm² or higher, the applicability date of the existing Catawba Unit 1 P-T limit curves decreased from 34 EFPY to 30.7 EFPY with the MUR power uprate. Similarly, the applicability date of the 51 EFPY P-T limit curves for Catawba Unit 1 decreased from 51 EFPY to 42.7 EFPY with the MUR power uprate. For more detailed conclusions, refer to Section 8.2 for Catawba Unit 1.

8.1 MUR POWER UPRATE ART CALCULATIONS

Table 8.1-1 Calculation of the Catawba Unit 1 ART Values at the 1/4T Location for 34 EFPY

RV Material and Identification Number	R.G. 1.99, Rev. 2 Position	CF ^(a) (°F)	1/4T Fluence ^(b) ($\times 10^{19}$ n/cm ² , E > 1.0 MeV)	1/4T FF ^(b)	IRT _{NDT} ^(c) (°F)	ΔRT_{NDT} (°F)	σ_1 ^(c) (°F)	σ_Δ ^(d) (°F)	Margin (°F)	ART (°F)
US Forging 06	1.1	123.5	0.048	0.2870	-26	35.4	0	17.0	34.0	43
IS Forging 05	1.1	58	1.035	1.0096	-8	58.6	0	17.0	34.0	85
<i>Using credible surveillance data</i>	2.1	28.5	1.035	1.0096	-8	28.8	0	8.5	17.0	38
LS Forging 04	1.1	26	1.035	1.0096	-13	26.3	0	13.1	26.3	40
Bottom Head Ring 03	1.1	37	0.081	0.3752	14	13.9	0	6.9	13.9	42
US to IS Circ. Weld W06 (Heat # 899680)	1.1	41	0.048	0.2870	10	11.8	0	5.9	11.8	34
IS to LS Circ. Weld W05 (Heat # 895075)	1.1	54	1.035	1.0096	-51	54.5	0	27.3	54.5	58
<i>Using credible surveillance data</i>	2.1	28.5	1.035	1.0096	-51	28.8	0	14.0	28.0	6
LS to Bottom Head Ring Weld W04 (Heat # 899680)	1.1	41	0.081	0.3752	10	15.4	0	7.7	15.4	41

Notes:

- (a) Taken from Table 5-2 of this report.
- (b) 1/4T fluence and FF were calculated using Regulatory Guide 1.99, Revision 2, and the Catawba Unit 1 RV beltline wall thickness of 8.465 inches.
- (c) Initial RT_{NDT} values are measured and are taken from Table 3-1 of this report.
- (d) Per Appendix A of this report, the surveillance data of the forging and weld were deemed credible. Per the guidance of Regulatory Guide 1.99, Revision 2, the base metal $\sigma_\Delta = 17^\circ\text{F}$ for Position 1.1 and, with credible surveillance data, $\sigma_\Delta = 8.5^\circ\text{F}$ for Position 2.1; the weld metal $\sigma_\Delta = 28^\circ\text{F}$ for Position 1.1 and, with credible surveillance data, $\sigma_\Delta = 14^\circ\text{F}$ for Position 2.1. However, σ_Δ need not exceed $0.5 \cdot \Delta RT_{NDT}$.

Table 8.1-2 Calculation of the Catawba Unit 1 ART Values at the 3/4T Location for 34 EFPY

RV Material and Identification Number	R.G. 1.99, Rev. 2 Position	CF ^(a) (°F)	3/4T Fluence ^(b) ($\times 10^{19}$ n/cm ² , E > 1.0 MeV)	3/4T FF ^(b)	IRT _{NDT} ^(c) (°F)	Δ RT _{NDT} (°F)	σ_I ^(c) (°F)	σ_A ^(d) (°F)	Margin (°F)	ART (°F)
US Forging 06	1.1	123.5	0.017	0.1580	-26	19.5	0	9.8	19.5	13
IS Forging 05	1.1	58	0.375	0.7286	-8	42.3	0	17.0	34.0	68
<i>Using credible surveillance data</i>	2.1	28.5	0.375	0.7286	-8	20.8	0	8.5	17.0	30
LS Forging 04	1.1	26	0.375	0.7286	-13	18.9	0	9.5	18.9	25
Bottom Head Ring 03	1.1	37	0.029	0.2162	14	8.0	0	4.0	8.0	30
US to IS Circ. Weld W06 (Heat # 899680)	1.1	41	0.017	0.1580	10	6.5	0	3.2	6.5	23
IS to LS Circ. Weld W05 (Heat # 895075)	1.1	54	0.375	0.7286	-51	39.3	0	19.7	39.3	28
<i>Using credible surveillance data</i>	2.1	28.5	0.375	0.7286	-51	20.8	0	10.4	20.8	-9
LS to Bottom Head Ring Weld W04 (Heat # 899680)	1.1	41	0.029	0.2162	10	8.9	0	4.4	8.9	28
Notes: (a) Taken from Table 5-2 of this report. (b) 3/4T fluence and FF were calculated using Regulatory Guide 1.99, Revision 2, and the Catawba Unit 1 RV beltline wall thickness of 8.465 inches. (c) Initial RT _{NDT} values are measured and are taken from Table 3-1 of this report. (d) Per Appendix A of this report, the surveillance data of the forging and weld were deemed <u>credible</u> . Per the guidance of Regulatory Guide 1.99, Revision 2, the base metal $\sigma_A = 17^\circ\text{F}$ for Position 1.1 and, with <u>credible</u> surveillance data, $\sigma_A = 8.5^\circ\text{F}$ for Position 2.1; the weld metal $\sigma_A = 28^\circ\text{F}$ for Position 1.1 and, with <u>credible</u> surveillance data, $\sigma_A = 14^\circ\text{F}$ for Position 2.1. However, σ_A need not exceed $0.5 \cdot \Delta\text{RT}_{\text{NDT}}$.										

Table 8.1-3 Calculation of the Catawba Unit 1 ART Values at the 1/4T Location for 54 EFPY

RV Material and Identification Number	R.G. 1.99, Rev. 2 Position	CF ^(a) (°F)	1/4T Fluence ^(b) ($\times 10^{19}$ n/cm ² , E > 1.0 MeV)	1/4T FF ^(b)	IRT _{NDT} ^(c) (°F)	Δ RT _{NDT} (°F)	σ_1 ^(c) (°F)	σ_Δ ^(d) (°F)	Margin (°F)	ART (°F)
US Forging 06	1.1	123.5	0.070	0.3488	-26	43.1	0	17.0	34.0	51
IS Forging 05	1.1	58	1.565	1.1237	-8	65.2	0	17.0	34.0	91
<i>Using credible surveillance data</i>	2.1	28.5	1.565	1.1237	-8	32.0	0	8.5	17.0	41
LS Forging 04	1.1	26	1.565	1.1237	-13	29.2	0	14.6	29.2	45
Bottom Head Ring 03	1.1	37	0.117	0.4496	14	16.6	0	8.3	16.6	47
US to IS Circ. Weld W06 (Heat # 899680)	1.1	41	0.070	0.3488	10	14.3	0	7.2	14.3	39
IS to LS Circ. Weld W05 (Heat # 895075)	1.1	54	1.565	1.1237	-51	60.7	0	28.0	56.0	66
<i>Using credible surveillance data</i>	2.1	28.5	1.565	1.1237	-51	32.0	0	14.0	28.0	9
LS to Bottom Head Ring Weld W04 (Heat # 899680)	1.1	41	0.117	0.4496	10	18.4	0	9.2	18.4	47

Notes:

- (a) Taken from Table 5-2 of this report.
- (b) 1/4T fluence and FF were calculated using Regulatory Guide 1.99, Revision 2, and the Catawba Unit 1 RV beltline wall thickness of 8.465 inches.
- (c) Initial RT_{NDT} values are measured and are taken from Table 3-1 of this report.
- (d) Per Appendix A of this report, the surveillance data of the forging and weld were deemed credible. Per the guidance of Regulatory Guide 1.99, Revision 2, the base metal $\sigma_\Delta = 17^\circ\text{F}$ for Position 1.1 and, with credible surveillance data, $\sigma_\Delta = 8.5^\circ\text{F}$ for Position 2.1; the weld metal $\sigma_\Delta = 28^\circ\text{F}$ for Position 1.1 and, with credible surveillance data, $\sigma_\Delta = 14^\circ\text{F}$ for Position 2.1. However, σ_Δ need not exceed $0.5 \cdot \Delta\text{RT}_{\text{NDT}}$.

Table 8.1-4 Calculation of the Catawba Unit 1 ART Values at the 3/4T Location for 54 EFPY

RV Material and Identification Number	R.G. 1.99, Rev. 2 Position	CF ^(a) (°F)	3/4T Fluence ^(b) ($\times 10^{19}$ n/cm ² , E > 1.0 MeV)	3/4T FF ^(b)	IRT _{NDT} ^(c) (°F)	Δ RT _{NDT} (°F)	σ_I ^(c) (°F)	σ_Δ ^(d) (°F)	Margin (°F)	ART (°F)
US Forging 06	1.1	123.5	0.025	0.1984	-26	24.5	0	12.3	24.5	23
IS Forging 05	1.1	58	0.567	0.8410	-8	48.8	0	17.0	34.0	75
<i>Using credible surveillance data</i>	2.1	28.5	0.567	0.8410	-8	24.0	0	8.5	17.0	33
LS Forging 04	1.1	26	0.567	0.8410	-13	21.9	0	10.9	21.9	31
Bottom Head Ring 03	1.1	37	0.042	0.2678	14	9.9	0	5.0	9.9	34
US to IS Circ. Weld W06 (Heat # 899680)	1.1	41	0.025	0.1984	10	8.1	0	4.1	8.1	26
IS to LS Circ. Weld W05 (Heat # 895075)	1.1	54	0.567	0.8410	-51	45.4	0	22.7	45.4	40
<i>Using credible surveillance data</i>	2.1	28.5	0.567	0.8410	-51	24.0	0	12.0	24.0	-3
LS to Bottom Head Ring Weld W04 (Heat # 899680)	1.1	41	0.042	0.2678	10	11.0	0	5.5	11.0	32
Notes: (a) Taken from Table 5-2 of this report. (b) 3/4T fluence and FF were calculated using Regulatory Guide 1.99, Revision 2, and the Catawba Unit 1 RV beltline wall thickness of 8.465 inches. (c) Initial RT _{NDT} values are measured and are taken from Table 3-1 of this report. (d) Per Appendix A of this report, the surveillance data of the forging and weld were deemed <u>credible</u> . Per the guidance of Regulatory Guide 1.99, Revision 2, the base metal $\sigma_\Delta = 17^\circ\text{F}$ for Position 1.1 and, with <u>credible</u> surveillance data, $\sigma_\Delta = 8.5^\circ\text{F}$ for Position 2.1; the weld metal $\sigma_\Delta = 28^\circ\text{F}$ for Position 1.1 and, with <u>credible</u> surveillance data, $\sigma_\Delta = 14^\circ\text{F}$ for Position 2.1. However, σ_Δ need not exceed $0.5 \cdot \Delta\text{RT}_{\text{NDT}}$.										

8.2 P-T LIMIT CURVES APPLICABILITY EVALUATION

Tables 8.1-1 through 8.1-4 summarize the 1/4T and 3/4T ART calculations for Catawba Unit 1 at 34 and 54 EFPY.

The applicability of the existing Catawba Unit 1 34 EFPY P-T limit curves, contained in WCAP-15203, Revision 1 (Reference 15) and 51 EFPY P-T limit curves, contained in WCAP-15448, Revision 1 (Reference 22) is evaluated by comparing the updated ART values contained in Section 8.1.1 with those used in the References 15 and 22 calculations. The existing P-T limit curves for Catawba Unit 1 are based on the limiting beltline material ART values, which are influenced by both fluence and initial material properties of that material. Using the MUR power uprate fluence projections, the 1/4T and 3/4T ART values were recalculated in Tables 8.1-1 through 8.1-4 as part of this applicability evaluation for Catawba Unit 1. Since the capsule fluence values were also revised as part of the MUR power uprate, the Position 2.1 chemistry factor values were updated in Section 5 of this report. The comparison of limiting ART values is contained in Table 8.2-1 for the existing Catawba Unit 1 34 EFPY P-T limit curves, and Table 8.2-2 for the existing 51 EFPY P-T limit curves.

Table 8.2-1 Summary of the Catawba Unit 1 Limiting ART Values used in the Applicability Evaluation of the Existing 34 EFPY RV Heatup and Cooldown Curves

	1/4T Location		3/4T Location	
	Existing 34 EFPY Curves documented in WCAP-15203, Revision 1	MUR Uprate Evaluation at 34 EFPY (Table 8.1-1)	Existing 34 EFPY Curves documented in WCAP-15203, Revision 1	MUR Uprate Evaluation at 34 EFPY (Table 8.1-2)
Limiting ART (°F)	42	43	31	30
Limiting Material	LS Forging 04	US Forging 06	IS Forging 05 (using credible surveillance data)	IS Forging 05 (using credible surveillance data) & Bottom Head Ring 03

Table 8.2-2 Summary of the Catawba Unit 1 Limiting ART Values used in the Applicability Evaluation of the Existing 51 EFPY RV Heatup and Cooldown Curves

	1/4T Location		3/4T Location	
	Existing 51 EFPY Curves documented in WCAP-15448, Revision 1	MUR Uprate Evaluation at 54 EFPY (Table 8.1-3)	Existing 51 EFPY Curves documented in WCAP-15448, Revision 1	MUR Uprate Evaluation at 54 EFPY (Table 8.1-4)
Limiting ART (°F)	47	51	34	34
Limiting Material	LS Forging 04	US Forging 06	IS Forging 05 (using credible surveillance data)	Bottom Head Ring 03

34 EFPY P-T Limit Curves

Table 8.2-1 compares the MUR power uprate limiting ART values at 34 EFPY to the limiting ART values used in development of the existing 34 EFPY P-T limit curves that are documented in WCAP-15203, Revision 1 (Reference 15). The limiting ART values used to develop the existing P-T limit curves are documented in Table 10 of Reference 15.

Based on the comparison of the ART values in Table 8.2-1, the limiting 1/4T ART value (42°F) used in the development of the current P-T limit curves at 34 EFPY is slightly less than the MUR power uprate limiting 1/4T ART value (43°F) at 34 EFPY for Catawba Unit 1. However, the limiting 3/4T ART value (30°F) calculated for the MUR evaluation is bounded by the limiting 3/4T value (31°F) used in the development of the current P-T limit curves, as shown in Table 8.2-1. Note that due to the consideration of all materials projected to achieve fluence levels of 1×10^{17} n/cm² or higher at 34 EFPY, the limiting materials have changed since the existing 34 EFPY P-T limit curves were developed.

Using the MUR fluence values, the fluence in which the limiting material at the 1/4T location (Upper Shell Forging 06) would have an associated ART value of 42°F was calculated for Catawba Unit 1. This fluence was used to determine a revised EFPY for the current P-T limit curves by linearly interpolating the MUR uprate fluence projections in Table 2-2 of this report. With consideration of all RV materials that are projected to achieve surface fluence levels of 1×10^{17} n/cm² or higher at 34 EFPY, the applicability date for which the current heatup and cooldown P-T limit curves were developed decreased from 34 EFPY to 30.7 EFPY with the MUR power uprate for Catawba Unit 1.

51 EFPY P-T Limit Curves

Table 8.2-2 compares the MUR power uprate limiting ART values at 54 EFPY to the limiting ART values used in development of the existing 51 EFPY P-T limit curves that are documented in WCAP-15448, Revision 1 (Reference 22). The limiting ART values used to develop the existing P-T limit curves are documented in Table 10 of Reference 22.

Based on the comparison of the ART values in Table 8.2-2, the limiting 1/4T ART value (47°F) used in the development of the current P-T limit curves at 51 EFPY is less than the MUR power uprate limiting 1/4T ART value (51°F) at 54 EFPY for Catawba Unit 1. However, the limiting 3/4T ART value (34°F) calculated for the MUR evaluation is equal to the limiting 3/4T value (34°F) used in the development of the current P-T limit curves, as shown in Table 8.2-2. Note that due to the consideration of all materials projected to achieve fluence levels of 1×10^{17} n/cm² or higher at 51 EFPY, the limiting materials have changed since the existing 51 EFPY P-T limit curves were developed.

Using the MUR fluence values, the fluence in which the limiting material at the 1/4T location (Upper Shell Forging 06) would have an associated ART value of 47°F was calculated for Catawba Unit 1. This fluence was used to determine a revised EFPY for the current P-T limit curves by linearly interpolating the MUR uprate fluence projections in Table 2-2 of this report. With consideration of all RV materials that are projected to achieve surface fluence levels of 1×10^{17} n/cm² or higher at 51 EFPY, the applicability date for which the current heatup and cooldown P-T limit curves were developed decreased from 51 EFPY to 42.7 EFPY with the MUR power uprate for Catawba Unit 1.

P-T Limits Applicability Conclusion

With consideration of all RV materials that are projected to achieve fluence levels of 1×10^{17} n/cm² or higher, it is concluded that the MUR uprate evaluation does require a reduction of the existing Catawba Unit 1 P-T limit curves applicability dates. The revised applicability date of the 34 EFPY P-T limit curves decreases to 30.7 EFPY. Similarly, the revised applicability date of the 51 EFPY P-T limit curves decreases to 42.7 EFPY.

The new MUR uprate applicability dates are as follows:

Catawba Unit 1 – WCAP-15203, Revision 1 (Reference 15)

Current applicability: 34 EFPY

Revised applicability using MUR uprate fluence: 30.7 EFPY

Catawba Unit 1 – WCAP-15448, Revision 1 (Reference 22)

Current applicability: 51 EFPY

Revised applicability using MUR uprate fluence: 42.7 EFPY

9 SURVEILLANCE CAPSULE WITHDRAWAL SUMMARY

Table 9-1 summarizes the removal of the six surveillance capsules from the Catawba Unit 1 RV, meeting the requirements of ASTM E185-82 (Reference 4), as required by 10 CFR 50, Appendix H (Reference 23).

Table 9-1 Catawba Unit 1 Surveillance Capsule Withdrawal Summary				
Capsule	Capsule Location	Lead Factor^(a)	Withdrawal EFPY^(b)	Fluence^(a) ($\times 10^{19}$ n/cm², E > 1.0 MeV)
Z	301.5°	3.85	0.79	0.292
Y	241°	3.73	4.98	1.31
V	61°	3.72	9.29	2.31
X ^(c)	238.5°	3.88	9.29	2.41
U ^(c)	58.5°	3.88	9.29	2.41
W ^(d)	121.5°	4.00	14.69	3.51

Notes:

- (a) Updated as part of the MUR power uprate fluence evaluation.
- (b) EFPY from plant startup. The fluence evaluation supporting this effort did not consider 0.11 EFPY of neutron exposure resulting from pre-commercial operation of Catawba Unit 1, between January 1985 and June 1985. The impact of this omission on the fluence evaluation results has been assessed to be negligible, and the results remain valid within the 20% uncertainty criterion for fluence calculations.
- (c) Capsules X and U were removed from the RV at 9.29 EFPY and the dosimeters were tested. The material specimens were not tested and are being stored for potential future testing or further irradiation.
- (d) Capsule W was removed from the RV at 14.69 EFPY. This capsule was placed in the spent fuel pool following removal. The removed and untested specimens may either be tested or re-inserted into the RV to be further irradiated.

Based on the limiting ΔRT_{NDT} value (55.2°F) for Catawba Unit 1 documented in Table 6-1 of this report, Catawba Unit 1 is required to withdraw three surveillance capsules, with the third capsule able to be held without testing following withdrawal. To date, three capsules (Z, Y, and V) were withdrawn and tested per ASTM E185-82. Two capsules (X and U) were withdrawn and the dosimeters were tested, but the material specimens were not tested and are being stored for potential future testing or further irradiation. The last capsule (W) was withdrawn and placed in the spent fuel pool following removal.

The withdrawal schedule for these surveillance capsules meets the current recommendations of ASTM E185-82 as required by 10 CFR Part 50, Appendix H for a license extension through 60 years of operation, with consideration of the MUR power uprate. Furthermore, although Catawba Unit 1 does not have any capsules remaining in the RV, an EVND program is in place, meeting the requirements of Section XI.M31 of the GALL Report (Reference 11). If necessary in the future, the removed and untested specimens may either be tested or re-inserted into the RV to be further irradiated. Decisions regarding the stored specimens should be made when Duke Energy seeks to obtain a second 20-year extension to 80 years of plant operation.

10 REFERENCES

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APPENDIX A

SURVEILLANCE DATA CREDIBILITY EVALUATION

Introduction

Regulatory Guide 1.99, Revision 2 (Reference A-1) describes general procedures acceptable to the NRC staff for calculating the effects of neutron radiation embrittlement of the low-alloy steels currently used for light-water-cooled reactor vessels. Position C.2 of Regulatory Guide 1.99, Revision 2, describes the method for calculating the ART and Charpy upper-shelf energy of reactor vessel beltline materials using surveillance capsule data. The methods of Position C.2 can only be applied when two or more credible surveillance data sets become available from the reactor in question.

To date there have been three surveillance capsules removed and tested from the Catawba Unit 1 reactor vessel. To use these surveillance data sets, they must be shown to be credible. In accordance with Regulatory Guide 1.99, Revision 2, the credibility of the surveillance data will be judged based on five criteria.

The purpose of this evaluation is to apply the credibility requirements of Regulatory Guide 1.99, Revision 2, to the Catawba Unit 1 RV surveillance data and determine if that surveillance data is credible.

Evaluation

Criterion 1: Materials in the capsules should be those judged most likely to be controlling with regard to radiation embrittlement.

The beltline region of the reactor vessel is defined in Appendix G to 10 CFR Part 50, "Fracture Toughness Requirements," (Reference A-2) as follows:

"the region of the reactor vessel (shell material including welds, heat affected zones, and plates or forgings) that directly surrounds the effective height of the active core and adjacent regions of the reactor vessel that are predicted to experience sufficient neutron radiation damage to be considered in the selection of the most limiting material with regard to radiation damage."

The Catawba Unit 1 reactor vessel beltline region consists of the following materials:

1. Intermediate Shell Forging 05
2. Lower Shell Forging 04
3. Intermediate Shell Forging to Lower Shell Forging Circumferential Weld Seam W05 (Weld Wire Heat # 895075, Flux Type Grau L.O. # LW320, Flux Lot # P46)

The Catawba Unit 1 surveillance program utilizes axial and tangential test specimens from Intermediate Shell Forging 05. The surveillance weld metal was fabricated with weld wire Heat # 895075, Flux Type Grau L.O., Flux Lot # P46.

Intermediate Shell Forging 05 had the highest initial RT_{NDT} and highest weight percent copper out of the two beltline forgings in the Catawba Unit 1 reactor vessel. Thus, it was selected as the surveillance base metal.

The weld material in the Catawba Unit 1 surveillance program was made of the same material as the reactor vessel beltline circumferential weld. In accordance with the definition of the reactor vessel beltline at that time, this was the only weld in the beltline region.

Therefore, the materials selected for use in the Catawba Unit 1 surveillance program were those judged to be most likely limiting with regard to radiation embrittlement according to the accepted methodology at the time the surveillance program was developed.

Based on the discussion, Criterion 1 is met for the Catawba Unit 1 surveillance program.

Criterion 2: Scatter in the plots of Charpy energy versus temperature for the irradiated and unirradiated conditions should be small enough to permit the determination of the 30 ft-lb temperature and upper-shelf energy unambiguously.

Plots of Charpy energy versus temperature for the unirradiated and irradiated conditions are presented in Section 5 and Appendix C of the latest surveillance capsule report, WCAP-15117 (Reference A-3).

Based on engineering judgment, the scatter in the data presented in these plots is small enough to permit the determination of the 30 ft-lb temperature and the upper-shelf energy of the Catawba Unit 1 surveillance materials unambiguously.

Hence, Criterion 2 is met for the Catawba Unit 1 surveillance program.

Criterion 3: When there are two or more sets of surveillance data from one reactor, the scatter of ΔRT_{NDT} values about a best-fit line drawn as described in Regulatory Position 2.1 normally should be less than 28°F for welds and 17°F for base metal. Even if the fluence range is large (two or more orders of magnitude), the scatter should not exceed twice those values. Even if the data fail this criterion for use in shift calculations, they may be credible for determining decrease in upper-shelf energy if the upper shelf can be clearly determined, following the definition given in ASTM E185-82 (Reference A-4).

The functional form of the least-squares method as described in Regulatory Position 2.1 will be utilized to determine a best-fit line for this data and to determine if the scatter of these ΔRT_{NDT} values about this line is less than 28°F for the weld and less than 17°F for the forging.

Following is the calculation of the best-fit line as described in Regulatory Position 2.1 of Regulatory Guide 1.99, Revision 2. In addition, the recommended NRC methods for determining credibility will be followed. The NRC methods were presented to industry at a meeting held by the NRC on February 12 and 13, 1998 (Reference A-5). At this meeting the NRC presented five cases. Of the five cases, Case 1 ("Surveillance data available from plant but no other source") most closely represents the situation for the Catawba Unit 1 surveillance forging material. However, Catawba Unit 1 has a weld that will be evaluated for credibility using the guidance for the appropriate case as explained in Reference A-5. Weld Heat # 895075 pertains to IS to LS circumferential weld W05 in the Catawba Unit 1 reactor vessel. This weld heat is contained in the Catawba Unit 1 surveillance program as well as the McGuire Unit 2 and Watts Bar Unit 1 surveillance programs. NRC Case 4 per Reference A-5 is entitled "Surveillance Data from Plant and Other Sources" and most closely represents the situation for Catawba Unit 1 weld Heat # 895075.

Case 1: IS Forging 05 andCase 4: Weld Heat # 895075 (Catawba Unit 1 data only)

Following the NRC Case 4 guidelines, the Catawba Unit 1 surveillance weld metal (Heat # 895075) will be evaluated first using Catawba Unit 1 data only. The Catawba Unit 1 surveillance forging data will also be evaluated here since only Catawba Unit 1 data are considered. Table A-1 contains these evaluations.

Table A-1 Calculation of Interim Chemistry Factors for the Credibility Evaluation Using Catawba Unit 1 Surveillance Capsule Data Only						
Material	Capsule	Capsule f ^(a) (x10 ¹⁹ n/cm ² , E > 1.0 MeV)	FF ^(b)	ΔRT _{NDT} ^(c) (°F)	FF*ΔRT _{NDT} (°F)	FF ²
IS Forging 05 (Axial)	Z	0.292	0.663	15.74	10.44	0.440
	Y	1.31	1.075	48.63	52.28	1.156
	V	2.31	1.226	50.58	62.03	1.504
IS Forging 05 (Tangential)	Z	0.292	0.663	0.0 ^(d)	0.00	0.440
	Y	1.31	1.075	19.09	20.52	1.156
	V	2.31	1.226	25.61	31.41	1.504
	SUM:				176.68	6.199
	CF _{IS Forging} = Σ(FF * ΔRT _{NDT}) ÷ Σ(FF ²) = (176.68) ÷ (6.199) = 28.5°F					
Surveillance Weld Material (Heat # 895075)	Z	0.292	0.663	1.91	1.27	0.440
	Y	1.31	1.075	17.79	19.13	1.156
	V	2.31	1.226	26.5	32.50	1.504
	SUM:				52.89	3.100
	CF _{Surv. Weld} = Σ(FF * ΔRT _{NDT}) ÷ Σ(FF ²) = (52.89) ÷ (3.100) = 17.1°F					
Notes:						
(a) f = capsule fluence taken from Table 4-1 of this report.						
(b) FF = fluence factor = f ^(0.28 - 0.10*log f) .						
(c) ΔRT _{NDT} values are the measured 30 ft-lb shift values taken from Table 4-1 of this report. These measured ΔRT _{NDT} values for the surveillance weld metal do not include the adjustment ratio procedure of Reg. Guide 1.99, Revision 2, Position 2.1, since this calculation is based on the actual surveillance weld metal measured shift values. In addition, only Catawba Unit 1 data are being considered; therefore, no temperature adjustment is required.						
(d) Original value was negative, but physically a reduction should not occur; therefore, a value of zero will be used.						

The scatter of ΔRT_{NDT} values about the functional form of a best-fit line drawn as described in Regulatory Position 2.1 is presented in Table A-2.

Material	Capsule	CF (Slope_{best-fit}) (°F)	Capsule f ($\times 10^{19}$ n/cm², E > 1.0 MeV)	FF	Measured ΔRT_{NDT} (°F)	Predicted $\Delta RT_{NDT}^{(a)}$ (°F)	Scatter $\Delta RT_{NDT}^{(b)}$ (°F)	<17°F (Base Metal) <28°F (Weld)
IS Forging 05 (Axial)	Z	28.5	0.292	0.663	15.74	18.9	3.2	Yes
	Y	28.5	1.31	1.075	48.63	30.6	18.0	No
	V	28.5	2.31	1.226	50.58	34.9	15.6	Yes
IS Forging 05 (Tangential)	Z	28.5	0.292	0.663	0.0	18.9	18.9	No
	Y	28.5	1.31	1.075	19.09	30.6	11.6	Yes
	V	28.5	2.31	1.226	25.61	34.9	9.3	Yes
Surveillance Weld Material (Heat # 895075)	Z	17.1	0.292	0.663	1.91	11.3	9.4	Yes
	Y	17.1	1.31	1.075	17.79	18.3	0.6	Yes
	V	17.1	2.31	1.226	26.5	20.9	5.6	Yes
Notes: (a) Predicted $\Delta RT_{NDT} = CF_{best-fit} * FF$. (b) Scatter $\Delta RT_{NDT} = \text{Absolute Value } [\text{Predicted } \Delta RT_{NDT} - \text{Measured } \Delta RT_{NDT}]$.								

The scatter of ΔRT_{NDT} values about the best-fit line, drawn as described in Regulatory Guide 1.99, Revision 2, Position 2.1, should be less than 17°F for base metal. Table A-2 indicates that two of the six surveillance data points fall outside the $\pm 1\sigma$ of 17°F scatter band for surveillance forging materials. From a statistical point of view, $\pm 1\sigma$ would be expected to encompass 68% of the data. Since 66.7% (two-thirds) of the forging data fall within the $\pm 1\sigma$ scatter band, it is concluded that this is approximately 68% and meets the intent of the requirement. Also, note that the two surveillance forging data points (Capsule Y axial and Capsule Z tangential) that fall outside the scatter band are only slightly outside the criteria by approximately 1°F or 2°F, respectively. The net effect of these deviations relative to the 1σ bounds is not considered to be statistically significant at a typical level of confidence. Therefore, based on engineering judgment, the IS Forging 05 surveillance data are deemed “credible” per the third criterion. Note that the data shown above in Table A-2 along with the credibility conclusions contained herein are consistent with the previous credibility evaluation contained Appendix D of Reference A-3.

The scatter of ΔRT_{NDT} values about the best-fit line, drawn as described in Regulatory Guide 1.99, Revision 2, Position 2.1, should be less than 28°F for weld metal. Table A-2 indicates that all three surveillance data points fall within the $\pm 1\sigma$ of 28°F scatter band for surveillance weld materials; therefore, the weld metal (Heat # 895075) is deemed “credible” per the third criterion when only the Catawba Unit 1 data are considered.

Case 4: Weld Heat # 895075 (All data)

In accordance with the NRC Case 4 guidelines, the data from Catawba Unit 1, McGuire Unit 2, and Watts Bar Unit 1 will now be analyzed together. Since the data are from multiple sources, the data are adjusted to the mean chemical composition and operating temperature of the surveillance capsules. This is performed in Table A-3 below.

Table A-3 Mean Chemical Composition and Operating Temperature for Catawba Unit 1, McGuire Unit 2, and Watts Bar Unit 1				
Material	Capsule	Cu Wt. %^(a)	Ni Wt. %^(a)	Plant Inlet Temperature^(b) (°F)
Weld Metal Heat # 895075 (Catawba Unit 1 data)	Z	0.05	0.73	553
	Y	0.05	0.73	553
	V	0.05	0.73	553
Weld Metal Heat # 895075 (McGuire Unit 2 data)	V	0.04	0.74	557
	X	0.04	0.74	557
	U	0.04	0.74	557
	W	0.04	0.74	557
Weld Metal Heat # 895075 (Watts Bar Unit 1 data)	U	0.03	0.75	560
	W	0.03	0.75	560
	X	0.03	0.75	560
	Z	0.03	0.75	560
MEAN		0.04	0.74	557.0
Notes: (a) All copper and nickel weight percent values are documented in Table 3-1 of this report, except for the calculated mean values. (b) All inlet temperature values are documented in Table 4-2 of this report, except for the calculated mean value.				

Therefore, the Catawba Unit 1, McGuire Unit 2, and Watts Bar Unit 1 surveillance capsule data will be adjusted to the mean chemical composition and operating temperature calculated in Table A-3.

Catawba Unit 1 data

$$CF_{\text{Mean}} = 54^{\circ}\text{F} \quad (\text{calculated per Table 1 of Regulatory Guide 1.99, Revision 2 using Cu Wt. \% = 0.04 and Ni Wt. \% = 0.74 per Table A-3})$$

$$CF_{\text{Surv. Weld (Catawba Unit 1)}} = 68^{\circ}\text{F} \quad (\text{per Table 5-2 of this report})$$

$$\text{Ratio} = 54 \div 68 = 0.79 \quad (\text{applied to Catawba Unit 1 surveillance data for weld Heat \# 895075 in the credibility evaluation})$$

McGuire Unit 2 data

$$CF_{\text{Mean}} = 54^{\circ}\text{F}$$

$$CF_{\text{Surv. Weld (McGuire Unit 2)}} = 54^{\circ}\text{F} \quad (\text{per Table 5-2 of this report})$$

$$\text{Ratio} = 54 \div 54 = 1.00 \quad (\text{no ratio is applied to McGuire Unit 2 surveillance data for weld Heat \# 895075 in the credibility evaluation since ratio} = 1.00)$$

Watts Bar Unit 1 data

$$CF_{\text{Mean}} = 54^{\circ}\text{F}$$

$$CF_{\text{Surv. Weld (Watts Bar Unit 1)}} = 41^{\circ}\text{F} \quad (\text{per Table 5-2 of this report})$$

$$\text{Ratio} = 54 \div 41 = 1.32 \quad (\text{applied to Watts Bar Unit 1 surveillance data for weld Heat \# 895075 in the credibility evaluation})$$

The capsule-specific temperature adjustments are as shown in Table A-4 below:

Table A-4 Operating Temperature Adjustments for the Catawba Unit 1, McGuire Unit 2, and Watts Bar Unit 1 Surveillance Capsule Data			
Material	Plant Inlet Temperature (°F)	Mean Operating Temperature (°F)	Temperature Adjustment (°F)
Weld Metal Heat # 895075 (Catawba Unit 1 data)	553	557.0	-4
Weld Metal Heat # 895075 (McGuire Unit 2 data)	557	557.0	0
Weld Metal Heat # 895075 (Watts Bar Unit 1 data)	560	557.0	+3

Using the chemical composition and operating temperature adjustments described and calculated previously, an interim chemistry factor is calculated for weld Heat # 895075 using the Catawba Unit 1, McGuire Unit 2, and Watts Bar Unit 1 data. This calculation is shown in Table A-5.

Table A-5 Calculation of Weld Heat # 895075 Interim Chemistry Factor for the Credibility Evaluation Using Catawba Unit 1, McGuire Unit 2, and Watts Bar Unit 1 Surveillance Capsule Data

Material	Capsule	Capsule $f^{(a)}$ ($\times 10^{19}$ n/cm ² , E > 1.0 MeV)	FF ^(b)	$\Delta RT_{NDT}^{(c)}$ (°F)	FF* ΔRT_{NDT} (°F)	FF ²
Catawba Unit 1 Surveillance Weld (Heat # 895075)	Z	0.292	0.663	-1.65 (1.91)	-1.10	0.440
	Y	1.31	1.075	10.89 (17.79)	11.71	1.156
	V	2.31	1.226	17.78 (26.5)	21.80	1.504
McGuire Unit 2 Surveillance Weld (Heat # 895075)	V	0.302	0.672	38.51 (38.51)	25.88	0.452
	X	1.38	1.089	35.93 (35.93)	39.14	1.187
	U	1.90	1.176	23.81 (23.81)	27.99	1.382
	W	2.82	1.276	43.76 (43.76)	55.83	1.628
Watts Bar Unit 1 Surveillance Weld (Heat # 895075)	U	0.447	0.776	3.96 (0.0 ^(d))	3.07	0.602
	W	1.08	1.022	44.22 (30.5)	45.17	1.044
	X	1.71	1.148	38.02 (25.8)	43.63	1.317
	Z	2.40	1.236	22.31 (13.9)	27.57	1.528
SUM:					300.71	12.238
$CF_{Heat \# 895075} = \Sigma(FF * \Delta RT_{NDT}) \div \Sigma(FF^2) = (300.71) \div (12.238) = 24.6^{\circ}F$						

Notes:

- (a) f = capsule fluence taken from Tables 4-1 and 4-2 of this report.
- (b) FF = fluence factor = $f^{(0.28 - 0.10 \log f)}$.
- (c) ΔRT_{NDT} values are the measured 30 ft-lb shift values. Pre-adjusted values are taken from Tables 4-1 and 4-2 of this report. ΔRT_{NDT} values are adjusted first by the difference in operating temperature then using the ratio procedure to account for differences between each surveillance weld's chemistry and the mean surveillance weld chemistry for Heat # 895075 (pre-adjusted values are listed in parentheses). The temperature adjustments are shown in Table A-4 of this report. The ratios applied are 0.79 for Catawba Unit 1 and 1.32 for Watts Bar Unit 1. No ratio is applied to the McGuire Unit 2 data since the ratio is equal to 1.
- (d) This ΔRT_{NDT} value was determined to be negative, but physically a reduction should not occur; therefore, a value of zero is used.

The scatter of ΔRT_{NDT} values about the functional form of a best-fit line drawn as described in Regulatory Position 2.1 is presented in Table A-6.

Table A-6 Best-Fit Evaluation for Surveillance Weld Metal Heat # 895075 Using Catawba Unit 1, McGuire Unit 2, and Watts Bar Unit 1 Data								
Material	Capsule	CF (Slope_{best-fit}) (°F)	Capsule f ($\times 10^{19}$ n/cm², E > 1.0 MeV)	FF	Measured ΔRT_{NDT} (°F)	Predicted $\Delta RT_{NDT}^{(a)}$ (°F)	Scatter $\Delta RT_{NDT}^{(b)}$ (°F)	<28°F (Weld)
Catawba Unit 1 Surveillance Weld (Heat # 895075)	Z	24.6	0.292	0.663	-1.65	16.3	18.0	Yes
	Y	24.6	1.31	1.075	10.89	26.4	15.5	Yes
	V	24.6	2.31	1.226	17.78	30.2	12.4	Yes
McGuire Unit 2 Surveillance Weld (Heat # 895075)	V	24.6	0.302	0.672	38.51	16.5	22.0	Yes
	X	24.6	1.38	1.089	35.93	26.8	9.2	Yes
	U	24.6	1.90	1.176	23.81	28.9	5.1	Yes
	W	24.6	2.82	1.276	43.76	31.4	12.4	Yes
Watts Bar Unit 1 Surveillance Weld (Heat # 895075)	U	24.6	0.447	0.776	3.96	19.1	15.1	Yes
	W	24.6	1.08	1.022	44.22	25.1	19.1	Yes
	X	24.6	1.71	1.148	38.02	28.2	9.8	Yes
	Z	24.6	2.40	1.236	22.31	30.4	8.1	Yes
Notes: (a) Predicted $\Delta RT_{NDT} = CF_{best-fit} * FF$. (b) Scatter $\Delta RT_{NDT} = \text{Absolute Value} [\text{Predicted } \Delta RT_{NDT} - \text{Measured } \Delta RT_{NDT}]$.								

The scatter of ΔRT_{NDT} values about the best-fit line, drawn as described in Regulatory Guide 1.99, Revision 2, Position 2.1, should be less than 28°F for weld metal. Table A-6 indicates that all eleven surveillance data points fall within the $\pm 1\sigma$ of 28°F scatter band for surveillance weld materials; therefore, the weld material (Heat # 895075) is deemed “credible” per the third criterion when all available data are considered.

In conclusion, the combined surveillance data from Catawba Unit 1, McGuire Unit 2, and Watts Bar Unit 1 for weld Heat # 895075 may be applied to the Catawba Unit 1 reactor vessel weld. The chemistry factor calculation as applicable to the McGuire Unit 2 reactor vessel weld is contained in Table 5-1 of this report.

Criterion 4: The irradiation temperature of the Charpy specimens in the capsule should match the vessel wall temperature at the cladding/base metal interface within +/- 25°F.

The Catawba Unit 1 capsule specimens are located in the reactor between the neutron pads and the vessel wall and are positioned opposite the center of the core. The test capsules are in baskets attached to the neutron pads. The location of the specimens with respect to the reactor vessel beltline provides assurance that the reactor vessel wall and the specimens experience equivalent operating conditions and will not differ by more than 25°F.

Hence, Criterion 4 is met for the Catawba Unit 1 surveillance program.

Criterion 5: The surveillance data for the correlation monitor material in the capsule should fall within the scatter band of the database for that material.

The Catawba Unit 1 surveillance program does not contain correlation monitor material; therefore, this criterion is not applicable to the Catawba Unit 1 surveillance program.

Conclusion

Based on the preceding responses to the five criteria of Regulatory Guide 1.99, Revision 2, Section B, and the application of engineering judgment, the Catawba Unit 1 surveillance forging data are deemed credible and the surveillance weld data are deemed credible when considering Catawba Unit 1 data only and also when considering all available data (Catawba Unit 1, McGuire Unit 2, and Watts Bar Unit 1).

Appendix A References

- A-1 U.S. Nuclear Regulatory Commission, Office of Nuclear Regulatory Research, Regulatory Guide 1.99, Revision 2, "Radiation Embrittlement of Reactor Vessel Materials," May 1988.
- A-2 10 CFR 50, Appendix G, "Fracture Toughness Requirements," Federal Register, Volume 60, No. 243, December 19, 1995.
- A-3 WCAP-15117, Revision 0, "Analysis of Capsule V and the Dosimeters from Capsules U and X from the Duke Power Company Catawba Unit 1 Reactor Vessel Radiation Surveillance Program," October 1998.
- A-4 ASTM E185-82, "Standard Practice for Conducting Surveillance Tests for Light-Water Cooled Nuclear Power Reactor Vessels," American Society of Testing and Materials, 1982.
- A-5 K. Wichman, M. Mitchell, and A. Hiser, USNRC, Generic Letter 92-01 and RPV Integrity Workshop Handouts, *NRC/Industry Workshop on RPV Integrity Issues*, February 12, 1998.

APPENDIX B

EMERGENCY RESPONSE GUIDELINE LIMITS

The Emergency Response Guideline (ERG) limits (Reference B-1) were developed in order to establish guidance for operator action in the event of an emergency situation, such as a PTS event. Generic categories of limits were developed for the guidelines based on the limiting inside surface RT_{NDT} . These generic categories were conservatively generated for the Westinghouse Owners Group (WOG) to be applicable to all Westinghouse plants.

The highest RT_{NDT} for which the generic category ERG limits were developed is 250°F for a longitudinal flaw and 300°F for a circumferential flaw. Therefore, if the limiting vessel material has an RT_{NDT} that exceeds 250°F for a longitudinal flaw or 300°F for a circumferential flaw, plant-specific ERG P-T limits must be developed.

The ERG category is determined by the magnitude of the limiting RT_{NDT} value, which is calculated the same way as the RT_{PTS} values are calculated in Section 6 of this report. The material with the highest RT_{NDT} defines the limiting material, which for Catawba Unit 1 is US Forging 06. Table B-1 identifies ERG category limits and the limiting material RT_{NDT} values at 54 EFPY for Catawba Unit 1.

Table B-1 Evaluation of Catawba Unit ERG Limit Category	
ERG Pressure-Temperature Limits (Reference B-1)	
Applicable RT_{NDT} Value ^(a)	ERG P-T Limit Category
$RT_{NDT} < 200^{\circ}\text{F}$	Category I
$200^{\circ}\text{F} < RT_{NDT} < 250^{\circ}\text{F}$	Category II
$250^{\circ}\text{F} < RT_{NDT} < 300^{\circ}\text{F}$	Category III b
Limiting RT_{NDT} Values	
Reactor Vessel Material	RT_{NDT} Value @ 54 EFPY
US Forging 06	63°F ^(b)
Notes: (a) Longitudinally oriented flaws are applicable only up to 250°F; circumferentially oriented flaws are applicable up to 300°F. (b) Value taken from Table 6-1 of this report.	

Emergency Response Guideline Limits Conclusion

Per Table B-1, the limiting material for Catawba Unit 1 (Upper Shell Forging 06) has an RT_{NDT} less than 200°F through 54 EFY. Therefore, Catawba Unit 1 remains in ERG Category I through EOLE (54 EFY).

Appendix B Reference

- B-1 HF04BG, "Background Information for Westinghouse Owners Group Emergency Response Guidelines, Critical Safety Function Status Tree, F-0.4 Integrity, HP/LP-Rev. 2," Westinghouse Owners Group, April 30, 2005.

APPENDIX C

VALIDATION OF THE RADIATION TRANSPORT MODELS BASED ON NEUTRON DOSIMETRY MEASUREMENTS

C.1 NEUTRON DOSIMETRY

Comparisons of measured dosimetry results to both the calculated and least-squares adjusted values for all surveillance capsules withdrawn from service to date at Catawba Unit 1 are described herein. Similarly, comparisons of measured EVND capsule results to both the calculated and least-squares adjusted values for EVND Capsules withdrawn from Catawba Unit 1 are also described in this section. The sensor sets from these capsules have been analyzed in accordance with the current dosimetry evaluation methodology described in Regulatory Guide 1.190, "Calculational and Dosimetry Methods for Determining Pressure Vessel Neutron Fluence" (Reference C-1). One of the main purposes for presenting this material is to demonstrate that the overall measurements agree with the calculated and least-squares adjusted values to within $\pm 20\%$ for in-vessel neutron dosimetry sensors and within $\pm 30\%$ for cavity (ex-vessel) neutron dosimetry sensors as specified by Regulatory Guide 1.190, thus serving to validate the calculated neutron exposures previously reported in Section 2.0 of this report.

C.1.1 Sensor Reaction Rate Determinations

In this section, the results of the evaluations of the five neutron sensor sets analyzed to date as part of the Catawba Unit 1 Reactor Vessel Materials Surveillance Program are presented. The capsule designation, location within the reactor, and time of withdrawal of each of these dosimetry sets were as follows:

Capsule ID	Azimuthal Location	Withdrawal Time	Irradiation Time [EFPY]
Z	31.5° Single	End of Cycle 1	0.79
Y	29.0° Dual	End of Cycle 6	4.98
V	29.0° Dual	End of Cycle 10	9.29
U	31.5° Dual	End of Cycle 10	9.29
X	31.5° Dual	End of Cycle 10	9.29

The passive neutron sensors included in the evaluations of Surveillance Capsules Z, Y, U, V, and X for Catawba Unit 1 are summarized as follows:

Sensor Material	Reaction Of Interest	Capsule Z	Capsule Y	Capsule V	Capsule U	Capsule X
Copper	$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	X	X	X	X	X
Iron	$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	X	X	X	X	X
Nickel	$^{58}\text{Ni}(n,p)^{58}\text{Co}$	X	X	X	X	X
Uranium-238	$^{238}\text{U}(n,f)^{137}\text{Cs}$	X	X	X	X	X
Neptunium-237	$^{237}\text{Np}(n,f)^{137}\text{Cs}$	X	X	X	X	X
Cobalt-Aluminum*	$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	X	X	X	X	X

Since all of the dosimetry monitors located at the radial center of the material test specimen array, radial gradient corrections were not required for these reaction rates. Pertinent physical and nuclear characteristics of the passive neutron sensors are listed in Table C-1.

The use of passive monitors such as those listed above does not yield a direct measure of the energy-dependent neutron flux at the point of interest. Rather, the activation or fission process is a measure of the integrated effect that the time and energy dependent neutron flux has on the target material over the course of the irradiation period. An accurate assessment of the average neutron flux level incident on the various monitors may be derived from the activation measurements only if the irradiation parameters are well known. In particular, the following variables are of interest:

1. the measured specific activity of each monitor,
2. the physical characteristics of each monitor,
3. the operating history of the reactor,
4. the energy response of each monitor, and
5. the neutron energy spectrum at the monitor location.

Results from the radiometric counting of the neutron sensors from Capsules Z, Y, V, U, and X are documented in References C-2 through C-4, respectively for Catawba Unit 1. Results from the radiometric counting of the neutron sensors from EVND Capsules A, B, C, D, E, and F are documented in Reference C-5. In all cases, the radiometric counting followed established ASTM procedures. Following sample preparation and weighing, the specific activity of each sensor was determined by means of a high-resolution gamma spectrometer. For the copper, iron, nickel, and cobalt-aluminum sensors, these analyses were performed by direct counting of each of the individual samples. In the case of the uranium and neptunium fission sensors, the analyses were carried out by direct counting preceded by dissolution and chemical separation of cesium from the sensor material.

The irradiation history of the reactor over the irradiation periods experienced by Capsules Z, Y, V, U, and X withdrawn from Catawba Unit 1 was based on the monthly power generation of Catawba Unit 1 from initial reactor commercial operation through the end of the dosimetry evaluation period. Please note the fluence evaluation supporting this effort did not consider 0.11 EFPY of neutron exposure resulting from pre-commercial operation of Catawba Unit 1, between January 1985 and June 1985. The impact of this

omission on the fluence evaluation results has been assessed to be negligible, and the results remain valid within the $\pm 20\%$ uncertainty criterion for fluence calculations. Future fluence evaluations will consider the pre-commercial operation phase of Catawba Unit 1. For the sensor sets utilized in the surveillance capsules, the half-lives of the product isotopes are long enough that a monthly histogram describing reactor operation has proven to be an adequate representation for use in radioactive decay corrections for the reactions of interest in the exposure evaluations. The irradiation history applicable to Capsules Z, Y, V, U, and X withdrawn from Catawba Unit 1 is given in Table 6-7 of Reference C-4.

Having the measured specific activities, the physical characteristics of the sensors, and the operating history of the reactor, reaction rates referenced to full-power operation were determined from the following equation:

$$R = \frac{A}{N_0 F Y \sum_j \frac{P_j}{P_{ref}} C_j [1 - e^{-\lambda t_j}] [e^{-\lambda t_{d,j}}]}$$

where:

R	=	Reaction rate averaged over the irradiation period and referenced to operation at a core power level of P_{ref} (rps/nucleus).
A	=	Measured specific activity (dps/g).
N_0	=	Number of target element atoms per gram of sensor.
F	=	Atom fraction of the target isotope in the target element.
Y	=	Number of product atoms produced per reaction.
P_j	=	Average core power level during irradiation period j (MW).
P_{ref}	=	Maximum or reference power level of the reactor (MW).
C_j	=	Calculated ratio of $\phi(E > 1.0 \text{ MeV})$ during irradiation period j to the time weighted average $\phi(E > 1.0 \text{ MeV})$ over the entire irradiation period.
λ	=	Decay constant of the product isotope (1/sec).
t_j	=	Length of irradiation period j (sec).
$t_{d,j}$	=	Decay time following irradiation period j (sec).

and the summation is carried out over the total number of monthly intervals comprising the irradiation period.

In the equation describing the reaction rate calculation, the ratio $[P_j]/[P_{ref}]$ accounts for month-by-month variation of reactor core power level within any given fuel cycle as well as over multiple fuel cycles. The ratio C_j , which was calculated for each fuel cycle using the transport methodology discussed in Section 2.0, accounts for the change in sensor reaction rates caused by variations in flux level induced by changes in core spatial power distributions from fuel cycle to fuel cycle. For a single-cycle irradiation, C_j is normally taken to be 1.0. However, for multiple-cycle irradiations, particularly those employing low leakage fuel management, the additional C_j term should be employed. The impact of changing flux levels for constant power operation can be quite significant for sensor sets that have been irradiated for many cycles in a reactor that has transitioned from non-low-leakage to low-leakage fuel management or for sensor sets contained in surveillance capsules that have been moved from one capsule location to another. The fuel cycle-specific neutron flux values along with the computed values for C_j are listed in Table C-2 for Catawba Unit 1. These flux values represent the cycle-dependent results at the radial and azimuthal center of the respective capsules at the axial elevation of the active fuel midplane.

Prior to using the measured reaction rates in the least-squares evaluations of the dosimetry sensor sets, additional corrections were made to the ^{238}U measurements to account for the presence of ^{235}U impurities in the sensors as well as to adjust for the build-in of plutonium isotopes over the course of the irradiation. Corrections were also made to the ^{238}U and ^{237}Np sensor reaction rates to account for gamma ray induced fission reactions that occurred over the course of the capsule irradiations. The correction factors applied to the Catawba Unit 1 fission sensor reaction rates are summarized as follows:

Correction	Capsule Z	Capsule Y	Capsule V	Capsule U	Capsule X
^{235}U Impurity/Pu Build-in	0.8728	0.8328	0.7991	0.7957	0.7957
$^{238}\text{U}(\gamma, f)$	0.9673	0.9680	0.9681	0.9669	0.9669
Net ^{238}U Correction	0.8443	0.8062	0.7736	0.7694	0.7694
$^{237}\text{Np}(\gamma, f)$	0.9903	0.9903	0.9903	0.9900	0.9900

These factors were applied in a multiplicative fashion to the decay-corrected uranium and neptunium fission sensor reaction rates.

Results of the sensor reaction rate determinations for Capsules Z, Y, V, U, and X from Catawba Unit 1 are given in Table C-3. In Table C-3, the measured specific activities, decay corrected saturated specific activities, and computed reaction rates for each sensor indexed to the radial center of the capsule are listed. The fission sensor reaction rates are listed both with and without the applied corrections for ^{238}U impurities, plutonium build-in, and gamma ray induced fission effects.

Results of the sensor reaction rate determinations for EVND Capsules A, B, C, D, E and F from Catawba Unit 1 are given in Table C-4. In Table C-4, the measured specific activities, decay corrected saturated specific activities, and computed reaction rates for each sensor indexed to the radial center of the capsule are listed. The fission sensor reaction rates are listed both with and without the applied corrections for ^{238}U impurities, plutonium build-in, and gamma ray induced fission effects.

C.1.2 Least-Squares Evaluation of Sensor Sets

Least-squares adjustment methods provide the capability of combining the measurement data with the corresponding neutron transport calculations resulting in a BE neutron energy spectrum with associated uncertainties. BEs for key exposure parameters such as $\phi(E > 1.0 \text{ MeV})$ or dpa/s along with their uncertainties are then easily obtained from the adjusted spectrum. In general, the least-squares methods, as applied to dosimetry evaluations, act to reconcile the measured sensor reaction rate data, dosimetry reaction cross sections, and the calculated neutron energy spectrum within their respective uncertainties. For example,

$$R_i \pm \delta_{R_i} = \sum_g (\sigma_{ig} \pm \delta_{\sigma_{ig}})(\phi_g \pm \delta_{\phi_g})$$

relates a set of measured reaction rates, R_i , to a single neutron spectrum, ϕ_g , through the multigroup dosimeter reaction cross section, σ_{ig} , each with an uncertainty δ . The primary objective of the least-squares evaluation is to produce unbiased estimates of the neutron exposure parameters at the location of the measurement.

For the least-squares evaluation of the Catawba Unit 1 surveillance capsule dosimetry and EVND, the FERRET code (Reference C-6) was employed to combine the results of the plant-specific neutron transport calculations and sensor set reaction rate measurements to determine BE values of exposure parameters ($\phi(E > 1.0 \text{ MeV})$ and dpa) along with associated uncertainties for the capsules analyzed to date.

The application of the least-squares methodology requires the following input:

1. The calculated neutron energy spectrum and associated uncertainties at the measurement location.
2. The measured reaction rates and associated uncertainty for each sensor contained in the multiple foil set.
3. The energy-dependent dosimetry reaction cross sections and associated uncertainties for each sensor contained in the multiple foil sensor set.

For the Catawba Unit 1 application, the calculated neutron spectrum was obtained from the results of plant-specific neutron transport calculations described in Section 2 of this report. The sensor reaction rates were derived from the measured specific activities using the procedures described in Section C.1.1. The dosimetry reaction cross sections and uncertainties were obtained from the SNLRML dosimetry cross-section library (Reference C-7). The SNLRML library is an evaluated dosimetry reaction cross-section compilation recommended for use in LWR evaluations by ASTM Standard E1018, "Application of ASTM Evaluated Cross-Section Data File, Matrix E 706 (IIB)."

The uncertainties associated with the measured reaction rates, dosimetry cross sections, and calculated neutron spectrum were input to the least-squares procedure in the form of variances and covariances. The assignment of the input uncertainties followed the guidance provided in ASTM Standard E 944, "Application of Neutron Spectrum Adjustment Methods in Reactor Surveillance."

The following provides a summary of the uncertainties associated with the least-squares evaluation of the Catawba Unit 1 surveillance capsule sensor sets.

Reaction Rate Uncertainties

The overall uncertainty associated with the measured reaction rates includes components due to the basic measurement process, irradiation history corrections, and corrections for competing reactions. A high level of accuracy in the reaction rate determinations is assured by utilizing laboratory procedures that conform to the ASTM National Consensus Standards for reaction rate determinations for each sensor type.

After combining all of these uncertainty components, the sensor reaction rates derived from the counting and data evaluation procedures were assigned the following net uncertainties for input to the least-squares evaluation:

Reaction	Uncertainty
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	5%
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	5%
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	5%
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	5%
$^{238}\text{U}(n,f)^{137}\text{Cs}$	10%
$^{237}\text{Np}(n,f)^{137}\text{Cs}$	10%
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	5%

These uncertainties are given at the 1σ level.

Dosimetry Cross-Section Uncertainties

The reaction rate cross sections used in the least-squares evaluations were taken from the SNLRML library. This data library provides reaction cross sections and associated uncertainties, including covariances, for 66 dosimetry sensors in common use. Both cross sections and uncertainties are provided in a fine multigroup structure for use in least-squares adjustment applications. These cross sections were compiled from the most recent cross-section evaluations and they have been tested with respect to their accuracy and consistency for least-squares evaluations. Further, the library has been empirically

tested for use in fission spectra determination as well as in the fluence and energy characterization of 14 MeV neutron sources.

For sensors included in the Catawba Unit 1 surveillance programs, the following uncertainties in the fission spectrum averaged cross sections are provided in the SNLRML documentation package.

Reaction	Uncertainty
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	4.51-4.87%
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	4.08-4.16%
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	3.05-3.11%
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	4.49-4.56%
$^{238}\text{U}(n,f)^{137}\text{Cs}$	0.54-0.64%
$^{237}\text{Np}(n,f)^{137}\text{Cs}$	10.32-10.97%
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	0.79-3.59%

These tabulated ranges provide an indication of the dosimetry cross-section uncertainties associated with the sensor sets used in LWR irradiations.

Calculated Neutron Spectrum

The neutron spectra input to the least-squares adjustment procedure were obtained directly from the results of plant-specific transport calculations for each surveillance capsule irradiation period and location. The spectrum for each capsule was input in an absolute sense (rather than as simply a relative spectral shape). Therefore, within the constraints of the assigned uncertainties, the calculated data were treated equally with the measurements.

While the uncertainties associated with the reaction rates were obtained from the measurement procedures and counting benchmarks and the dosimetry cross-section uncertainties were supplied directly with the SNLRML library, the uncertainty matrix for the calculated spectrum was constructed from the following relationship:

$$M_{gg'} = R_n^2 + R_g * R_{g'} * P_{gg'}$$

where R_n specifies an overall fractional normalization uncertainty and the fractional uncertainties R_g and $R_{g'}$ specify additional random groupwise uncertainties that are correlated with a correlation matrix given by:

$$P_{gg'} = [1 - \theta]\delta_{gg'} + \theta e^{-H}$$

where

$$H = \frac{(g - g')^2}{2\gamma^2}$$

The first term in the correlation matrix equation specifies purely random uncertainties, while the second term describes the short-range correlations over a group range γ (θ specifies the strength of the latter term). The value of δ is 1.0 when $g = g'$, and is 0.0 otherwise.

The set of parameters defining the input covariance matrix for the Catawba Unit 1 calculated spectra was as follows:

Flux Normalization Uncertainty (R_n)	15%
Flux Group Uncertainties ($R_g, R_{g'}$)	
($E > 0.0055$ MeV)	15%
(0.68 eV $< E < 0.0055$ MeV)	25%
($E < 0.68$ eV)	50%
Short Range Correlation (θ)	
($E > 0.0055$ MeV)	0.9
(0.68 eV $< E < 0.0055$ MeV)	0.5
($E < 0.68$ eV)	0.5
Flux Group Correlation Range (γ)	
($E > 0.0055$ MeV)	6
(0.68 eV $< E < 0.0055$ MeV)	3
($E < 0.68$ eV)	2

C.1.3 Comparisons of Measurements and Calculations

Results of the least-squares evaluations of the dosimetry from the Catawba Unit 1 surveillance capsules and EVND capsules withdrawn to date are provided in Tables C-5 through C-7, C-10, and C-11. In Tables C-5 and C-6, measured, calculated, and BE values for sensor reaction rates are given for each surveillance and EVND capsule. Also provided in this tabulation are ratios of the measured reaction rates to both the calculated and least-squares adjusted reaction rates. These ratios of measured/calculated (M/C) and measured/best estimate (M/BE) illustrate the consistency of the fit of the calculated neutron energy spectra to the measured reaction rates both before and after adjustment. In Tables C-7, C-10 and C-11, comparison of the calculated and BE values of neutron flux ($E > 1.0$ MeV) and iron atom displacement rate are tabulated along with the best-estimate/calculated (BE/C) ratios observed for each of the capsules.

The data comparisons provided in Tables C-5 through C-7, C-10, and C-11 show that the adjustments to the calculated spectra are relatively small and well within the assigned uncertainties for the calculated

spectra, measured sensor reaction rates, and dosimetry reaction cross sections. Further, these results indicate that the use of the least-squares evaluation results in a reduction in the uncertainties associated with the exposure of the surveillance capsules. From Section 2.0 of this report, it may be noted that the uncertainty associated with the unadjusted calculation of neutron fluence ($E > 1.0$ MeV) and iron atom displacements at the surveillance capsule locations is specified as 12% at the 1σ level. From Tables C-7, C-10, and C-11, it is noted that the corresponding uncertainties associated with the least-squares adjusted exposure parameters have been reduced to 6% for neutron flux ($E > 1.0$ MeV) and 8-9% for iron atom displacement rate. Again, the uncertainties from the least-squares evaluation are at the 1σ level.

Further comparisons of the measurement results (from Tables C-5 through C-7, C-10, and C-11) with calculations are given in Tables C-8 and C-9 for the surveillance capsules and Tables C-12 through C-15 for EVND capsules. These comparisons are given on two levels. In Tables C-8, C-12 and C-13, calculations of individual threshold sensor reaction rates are compared directly with the corresponding measurements. These threshold reaction rate comparisons provide a good evaluation of the accuracy of the fast neutron portion of the calculated energy spectra. In Tables C-9, C-14 and C-15, calculations of fast neutron exposure rates in terms of $\phi(E > 1.0$ MeV) and dpa/s are compared with the BE results obtained from the least-squares evaluation of the capsule dosimetry results. These two levels of comparison yield consistent and similar results with all measurement-to-calculation comparisons falling well within the 20% limits specified as the acceptance criteria in Regulatory Guide 1.190.

In the case of the direct comparison of measured and calculated sensor reaction rates, the M/C comparisons for fast neutron reactions range from 0.83 to 1.25 for the 25 samples included in the surveillance capsules data set. The overall average M/C ratio for the entire set of Catawba Unit 1 surveillance capsules data is 1.00 with an associated standard deviation of 12.8%. Similarly for Catawba Unit 1 EVND capsules at core midplane, the M/C comparisons for fast neutron reactions range from 0.81 to 1.10 for the 24 samples included in the data set. The overall average M/C ratio for the entire set of Catawba Unit 1 EVND capsules at core midplane data is 0.93 with an associated standard deviation of 8.1%. For Catawba Unit 1 EVND capsules off core midplane, the M/C comparisons for fast neutron reactions range from 0.69 to 1.29 for the 12 samples included in the data set. The overall average M/C ratio for the entire set of Catawba Unit 1 EVND capsules off core midplane data is 0.88 with an associated standard deviation of 19.0%.

In the comparisons of BE and calculated fast neutron exposure parameters for Catawba Unit 1 surveillance capsules, the corresponding BE/C comparisons for the capsule data sets range from 0.90 to 1.04 for neutron flux ($E > 1.0$ MeV) and from 0.92 to 1.06 for iron atom displacement rate. The overall average BE/C ratios for neutron flux ($E > 1.0$ MeV) and iron atom displacement rate are 0.96 with a standard deviation of 6.1% and 0.98 with a standard deviation of 6.4%, respectively. Similarly, for Catawba Unit 1 EVND capsules at core midplane, the corresponding BE/C comparisons for the capsule data sets range from 0.90 to 0.98 for neutron flux ($E > 1.0$ MeV) and from 0.92 to 0.98 for iron atom displacement rate. The overall average BE/C ratios for neutron flux ($E > 1.0$ MeV) and iron atom displacement rate are 0.93 with a standard deviation of 3.9% and 0.95 with a standard deviation of 3.2%, respectively. For Catawba Unit 1 EVND capsules off core midplane, the corresponding BE/C comparisons for the capsule data sets range from 0.79 to 1.02 for neutron flux ($E > 1.0$ MeV) and from

0.81 to 1.03 for iron atom displacement rate. The overall average BE/C ratios for neutron flux ($E > 1.0$ MeV) and iron atom displacement rate are 0.91 with a standard deviation of 18.0% and 0.92 with a standard deviation of 16.9%, respectively.

Based on these comparisons, it is concluded that the calculated fast neutron exposures provided in Section 2.0 of this report are validated for use in the assessment of the condition of the materials comprising the beltline region and extended beltline region of the Catawba Unit 1 reactor pressure vessel.

Table C-1 Nuclear Parameters Used in the Evaluation of Neutron Sensors

Monitor Material	Reaction of Interest	Target Atom Fraction	90% Response Range (MeV)	Product Half-life	Fission Yield (%)
Titanium	$^{46}\text{Ti} (n,p)$	0.0825	3.70 – 9.43	83.79 d	
Copper	$^{63}\text{Cu} (n,\alpha)$	0.6917	4.53– 11.0	5.271 y	
Iron	$^{54}\text{Fe} (n,p)$	0.0585	2.27 – 7.54	312.1 d	
Nickel	$^{58}\text{Ni} (n,p)$	0.6808	1.98 – 7.51	70.82 d	
Uranium-238	$^{238}\text{U} (n,f)$	1.0000	1.44 – 6.69	30.07 y	6.02
Neptunium-237	$^{237}\text{Np} (n,f)$	1.0000	0.68 – 5.61	30.07 y	6.17
Cobalt-Aluminum	$^{59}\text{Co} (n,\gamma)$	0.0015	non-threshold	5.271 y	

Note:

The 90% response range is defined such that, in the neutron spectrum characteristic of the Catawba Unit 1 surveillance capsules, approximately 90% of the sensor response is due to neutrons in the energy range specified with approximately 5% of the total response due to neutrons with energies below the lower limit and 5% of the total response due to neutrons with energies above the upper limit.

Table C-2 Calculated Fast Neutron Flux ($E > 1.0$ MeV) at Catawba Unit 1 Surveillance Capsule Center Core Midplane Elevation						
Fuel Cycle	Cycle Length [EFPS]	$\phi(E > 1.0 \text{ MeV})$ [n/cm²-s]				
		Capsule Z	Capsule Y	Capsule V	Capsule U	Capsule X
1	2.49E+07	1.17E+11	1.09E+11	1.09E+11	1.18E+11	1.18E+11
2	2.37E+07		7.68E+10	7.68E+10	8.27E+10	8.27E+10
3	2.43E+07		8.61E+10	8.61E+10	8.97E+10	8.97E+10
4	2.71E+07		7.75E+10	7.75E+10	7.99E+10	7.99E+10
5	2.49E+07		7.61E+10	7.61E+10	8.10E+10	8.10E+10
6	3.22E+07		7.83E+10	7.83E+10	7.81E+10	7.81E+10
7	3.00E+07			7.83E+10	8.25E+10	8.25E+10
8	3.38E+07			7.45E+10	7.42E+10	7.42E+10
9	3.69E+07			6.87E+10	7.03E+10	7.03E+10
10	3.53E+07			7.16E+10	7.74E+10	7.74E+10
Average		1.17E+11	8.37E+10	7.88E+10	8.23E+10	8.23E+10

Table C-2 Calculated C_j Factors at Catawba Unit 1 Surveillance Capsule Center Core Midplane Elevation (cont.)						
Fuel Cycle	Cycle Length [EFPS]	C_j				
		Capsule Z	Capsule Y	Capsule V	Capsule U	Capsule X
1	2.49E+07	1.00	1.30	1.38	1.43	1.43
2	2.37E+07		0.92	0.98	1.01	1.01
3	2.43E+07		1.03	1.09	1.09	1.09
4	2.71E+07		0.93	0.98	0.97	0.97
5	2.49E+07		0.91	0.97	0.98	0.98
6	3.22E+07		0.94	0.99	0.95	0.95
7	3.00E+07			0.99	1.00	1.00
8	3.38E+07			0.95	0.90	0.90
9	3.69E+07			0.87	0.85	0.85
10	3.53E+07			0.91	0.94	0.94
Average		1.00	1.00	1.00	1.00	1.00

Table C-3 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 Surveillance Capsule Z				
Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Top	4.39E+04	4.84E+05	7.38E-17
	Middle	4.46E+04	4.92E+05	7.50E-17
	Bottom	4.81E+04	5.30E+05	8.09E-17
	Average			7.66E-17
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Top	1.21E+06	4.34E+06	6.89E-15
	Bottom	1.32E+06	4.74E+06	7.51E-15
	Average			7.20E-15
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Top	6.17E+06	5.68E+07	8.13E-15
	Middle	7.34E+06	6.75E+07	9.67E-15
	Bottom	7.02E+06	6.46E+07	9.25E-15
	Average			9.01E-15
$^{238}\text{U} (n,f) ^{137}\text{Cs}$ (Cd)	Middle	1.44E+05	8.07E+06	5.30E-14
	Including ^{235}U , ^{239}Pu , and γ fission corrections:			4.47E-14
$^{237}\text{Np} (n,f) ^{137}\text{Cs}$ (Cd)	Middle	1.20E+06	6.72E+07	4.29E-13
	Including γ fission corrections:			4.25E-13
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Top	7.33E+06	8.08E+07	5.27E-12
	Middle	8.51E+06	9.38E+07	6.12E-12
	Bottom	8.99E+06	9.91E+07	6.47E-12
	N/A*	8.67E+06	9.56E+07	6.24E-12
	Average			6.03E-12
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$ (Cd)	Top	4.35E+06	4.80E+07	3.13E-12
	Middle	3.08E+06	3.40E+07	2.22E-12
	Bottom	4.29E+06	4.73E+07	3.09E-12
	Average			2.81E-12
Notes:				
1. Measured specific activities are indexed to a counting date of February 2, 1987.				
2. The average $^{238}\text{U} (n,f)$ reaction rate of 4.47E-14 includes a correction factor of 0.8728 to account for plutonium build-in and an additional factor of 0.9673 to account for photo-fission effects in the sensor.				
3. The average $^{237}\text{Np} (n,f)$ reaction rate of 4.25E-13 includes a correction factor of 0.9903 to account for photo-fission effects in the sensor.				

* The location of the dosimeter sensor cannot be determined from reference document.

Table C-3 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 Surveillance Capsule Y (cont.)

Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Top	1.41E+05	3.47E+05	5.29E-17
	Middle	1.29E+05	3.17E+05	4.84E-17
	Bottom	1.28E+05	3.15E+05	4.80E-17
	Average			4.98E-17
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Top	1.67E+06	3.00E+06	4.75E-15
	Bottom	1.55E+06	2.78E+06	4.41E-15
	Average			4.58E-15
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Top	1.12E+07	4.98E+07	7.13E-15
	Middle	1.04E+07	4.62E+07	6.62E-15
	Bottom	1.02E+07	4.53E+07	6.49E-15
	Average			6.74E-15
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	5.34E+05	5.09E+06	3.34E-14
	Including ^{235}U , ^{239}Pu , and γ fission corrections:			2.69E-14
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	4.27E+06	4.07E+07	2.60E-13
	Including γ fission corrections:			2.57E-13
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Top	2.07E+07	5.09E+07	3.32E-12
	Top	2.42E+07	5.95E+07	3.88E-12
	Middle	2.06E+07	5.07E+07	3.31E-12
	Middle	2.39E+07	5.88E+07	3.84E-12
	Bottom	2.33E+07	5.73E+07	3.74E-12
	Bottom	2.03E+07	4.99E+07	3.26E-12
	Average			3.56E-12
	Average			3.56E-12
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Top	1.28E+07	3.15E+07	2.05E-12
	Middle	1.29E+07	3.17E+07	2.07E-12
	Bottom	1.29E+07	3.17E+07	2.07E-12
	Average			2.07E-12

Notes:

1. Measured specific activities are indexed to a counting date of November 28, 1992.
2. The average $^{238}\text{U} (n,f)$ reaction rate of 2.69E-14 includes a correction factor of 0.8328 to account for plutonium build-in and an additional factor of 0.9680 to account for photo-fission effects in the sensor.
3. The average $^{237}\text{Np} (n,f)$ reaction rate of 2.57E-13 includes a correction factor of 0.9903 to account for photo-fission effects in the sensor.

Table C-3 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 Surveillance Capsule V (cont.)

Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Top	1.67E+05	2.98E+05	4.55E-17
	Middle	1.52E+05	2.72E+05	4.14E-17
	Bottom	1.49E+05	2.66E+05	4.06E-17
	Average			4.25E-17
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Top	1.36E+06	2.78E+06	4.41E-15
	Bottom	1.21E+06	2.48E+06	3.92E-15
	Average			4.17E-15
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Top	4.56E+06	4.19E+07	6.00E-15
	Middle	4.17E+06	3.83E+07	5.49E-15
	Bottom	4.10E+06	3.77E+07	5.39E-15
	Average			5.63E-15
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	9.84E+05	5.35E+06	3.52E-14
	Including ^{235}U , ^{239}Pu , and γ fission corrections:			2.72E-14
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	6.43E+06	3.50E+07	2.23E-13
	Including γ fission corrections:			2.21E-13
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Top	2.89E+07	5.16E+07	3.37E-12
	Top	2.75E+07	4.91E+07	3.21E-12
	Middle	2.42E+07	4.32E+07	2.82E-12
	Middle	2.85E+07	5.09E+07	3.32E-12
	Bottom	2.46E+07	4.39E+07	2.87E-12
	Bottom	2.82E+07	5.04E+07	3.29E-12
				3.14E-12
	Average			
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Top	1.52E+07	2.72E+07	1.77E-12
	Middle	1.40E+07	2.50E+07	1.63E-12
	Bottom	1.44E+07	2.57E+07	1.68E-12
	Average			1.69E-12

Notes:

1. Measured specific activities are indexed to a counting date of July 1, 1998.
2. The average $^{238}\text{U} (n,f)$ reaction rate of 2.72E-14 includes a correction factor of 0.7991 to account for plutonium build-in and an additional factor of 0.9681 to account for photo-fission effects in the sensor.
3. The average $^{237}\text{Np} (n,f)$ reaction rate of 2.21E-13 includes a correction factor of 0.9903 to account for photo-fission effects in the sensor.

Table C-3 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 Surveillance Capsule U (cont.)				
Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Top	1.78E+05	3.19E+05	4.87E-17
	Middle	1.60E+05	2.87E+05	4.37E-17
	Bottom	1.56E+05	2.80E+05	4.26E-17
	Average			4.50E-17
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Top	1.44E+06	2.90E+06	4.60E-15
	Middle	1.33E+06	2.68E+06	4.25E-15
	Bottom	1.30E+06	2.62E+06	4.15E-15
	Average			4.33E-15
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Top	4.95E+06	4.40E+07	6.30E-15
	Middle	4.60E+06	4.09E+07	5.85E-15
	Bottom	4.43E+06	3.94E+07	5.63E-15
	Average			5.93E-15
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	1.05E+06	5.72E+06	3.76E-14
	Including ^{235}U , ^{239}Pu , and γ fission corrections:			2.89E-14
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	6.51E+06	3.55E+07	2.26E-13
	Including γ fission corrections:			2.24E-13
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Top	3.05E+07	5.47E+07	3.57E-12
	Middle	2.91E+07	5.21E+07	3.40E-12
	Middle	2.54E+07	4.55E+07	2.97E-12
	Bottom	2.66E+07	4.77E+07	3.11E-12
	Bottom	2.31E+07	4.14E+07	2.70E-12
	Average			3.15E-12
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Top	1.58E+07	2.83E+07	1.85E-12
	Middle	1.56E+07	2.80E+07	1.82E-12
	Bottom	1.43E+07	2.56E+07	1.67E-12
	Average			1.78E-12
Notes:				
1. Measured specific activities are indexed to a counting date of July 1, 1998.				
2. The average $^{238}\text{U} (n,f)$ reaction rate of 2.89E-14 includes a correction factor of 0.7957 to account for plutonium build-in and an additional factor of 0.9669 to account for photo-fission effects in the sensor.				
3. The average $^{237}\text{Np} (n,f)$ reaction rate of 2.24E-13 includes a correction factor of 0.9900 to account for photo-fission effects in the sensor.				

Table C-3 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 Surveillance Capsule X (cont.)				
Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Top	1.74E+05	3.12E+05	4.76E-17
	Middle	1.65E+05	2.96E+05	4.51E-17
	Bottom	1.62E+05	2.90E+05	4.43E-17
	Average			4.57E-17
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Top	1.43E+06	2.88E+06	4.57E-15
	Middle	1.33E+06	2.68E+06	4.25E-15
	Bottom	1.33E+06	2.68E+06	4.25E-15
	Average			4.35E-15
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Top	5.04E+06	4.48E+07	6.41E-15
	Middle	4.68E+06	4.16E+07	5.95E-15
	Bottom	4.61E+06	4.10E+07	5.86E-15
	Average			6.08E-15
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	1.16E+06	6.32E+06	4.15E-14
	Including ^{235}U , ^{239}Pu , and γ fission corrections:			3.19E-14
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	8.56E+06	4.66E+07	2.97E-13
	Including γ fission corrections:			2.94E-13
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Top	3.12E+07	5.59E+07	3.65E-12
	Top	2.94E+07	5.27E+07	3.44E-12
	Middle	2.63E+07	4.71E+07	3.07E-12
	Middle	3.15E+07	5.64E+07	3.68E-12
	Bottom	3.08E+07	5.52E+07	3.60E-12
	Bottom	2.76E+07	4.95E+07	3.23E-12
	Average			3.44E-12
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Top	1.58E+07	2.83E+07	1.85E-12
	Middle	1.57E+07	2.81E+07	1.84E-12
	Bottom	1.58E+07	2.83E+07	1.85E-12
	Average			1.84E-12
Notes:				
1. Measured specific activities are indexed to a counting date of July 1, 1998.				
2. The average $^{238}\text{U} (n,f)$ reaction rate of 3.19E-14 includes a correction factor of 0.7957 to account for plutonium build-in and an additional factor of 0.9669 to account for photo-fission effects in the sensor.				
3. The average $^{237}\text{Np} (n,f)$ reaction rate of 2.94E-13 includes a correction factor of 0.9900 to account for photo-fission effects in the sensor.				

Table C-4 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 EVND Capsule A

Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{46}\text{Ti} (n,p) ^{46}\text{Sc}$	Middle	5.48E+02	5.07E+03	4.88E-18
	Average			4.88E-18
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Middle	7.82E+03	1.69E+04	2.69E-17
	Middle	7.94E+03	1.72E+04	2.73E-17
	Average			2.71E-17
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Middle	1.87E+04	2.53E+05	3.63E-17
	Average			3.63E-17
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Middle	6.32E+02	2.39E+03	3.64E-19
	Average			3.64E-19
$^{238}\text{U} (n,f) ^{95}\text{Zr} (\text{Cd})$	Middle	8.41E+02	1.55E+04	1.36E-16
		Including γ fission corrections:		1.30E-16
$^{238}\text{U} (n,f) ^{103}\text{Ru} (\text{Cd})$	Middle	1.60E+02	1.70E+04	1.22E-16
		Including γ fission corrections:		1.17E-16
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	9.74E+02	1.76E+04	1.31E-16
		Including γ fission corrections:		1.26E-16
$^{237}\text{Np} (n,f) ^{95}\text{Zr} (\text{Cd})$	Middle	1.34E+04	2.47E+05	1.96E-15
		Including γ fission corrections:		1.94E-15
$^{237}\text{Np} (n,f) ^{103}\text{Ru} (\text{Cd})$	Middle	2.35E+03	2.49E+05	2.02E-15
		Including γ fission corrections:		2.00E-15
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	1.51E+04	2.72E+05	1.99E-15
		Including γ fission corrections:		1.97E-15
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Middle	5.22E+05	1.97E+06	3.99E-14
	Average			3.99E-14
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Middle	2.45E+05	9.25E+05	1.87E-14
	Average			1.87E-14
Notes:				
1. Measured specific activities are indexed to a counting date of July 24, 2007.				
2. The average $^{238}\text{U} (n,f)$ reaction rates include correction factors to account for plutonium build-in and photo-fission effects in the sensor.				
3. The average $^{237}\text{Np} (n,f)$ reaction rates include correction factors to account for photo-fission effects in the sensor.				

Table C-4 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 EVND Capsule B (cont.)				
Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{46}\text{Ti} (n,p) ^{46}\text{Sc}$	Middle	7.89E+02	7.19E+03	6.93E-18
	Average			6.93E-18
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Middle	1.16E+04	2.49E+04	3.95E-17
	Middle	1.15E+04	2.47E+04	3.92E-17
	Average			3.93E-17
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Middle	2.78E+04	3.71E+05	5.31E-17
	Average			5.31E-17
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Middle	8.30E+02	3.13E+03	4.77E-19
	Average			4.77E-19
$^{238}\text{U} (n,f) ^{95}\text{Zr} (\text{Cd})$	Middle	1.26E+03	2.29E+04	2.00E-16
	Including γ fission corrections:			1.93E-16
$^{238}\text{U} (n,f) ^{103}\text{Ru} (\text{Cd})$	Middle	2.70E+02	2.82E+04	2.02E-16
	Including γ fission corrections:			1.95E-16
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	1.50E+03	2.71E+04	2.02E-16
	Including γ fission corrections:			1.95E-16
$^{237}\text{Np} (n,f) ^{95}\text{Zr} (\text{Cd})$	Middle	2.11E+04	3.84E+05	3.04E-15
	Including γ fission corrections:			3.02E-15
$^{237}\text{Np} (n,f) ^{103}\text{Ru} (\text{Cd})$	Middle	4.07E+03	4.25E+05	3.44E-15
	Including γ fission corrections:			3.42E-15
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	2.35E+04	4.24E+05	3.10E-15
	Including γ fission corrections:			3.08E-15
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Middle	7.60E+05	2.86E+06	5.80E-14
	Average			5.80E-14
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Middle	3.92E+05	1.48E+06	2.99E-14
	Average			2.99E-14
Notes:				
1. Measured specific activities are indexed to a counting date of July 24, 2007.				
2. The average $^{238}\text{U} (n,f)$ reaction rates include correction factors to account for plutonium build-in and photo-fission effects in the sensor.				
3. The average $^{237}\text{Np} (n,f)$ reaction rates include correction factors to account for photo-fission effects in the sensor.				

Table C-4 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 EVND Capsule C (cont.)				
Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{46}\text{Ti} (n,p) ^{46}\text{Sc}$	Middle	8.37E+02	7.27E+03	7.00E-18
	Average			7.00E-18
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Middle	1.29E+04	2.70E+04	4.28E-17
	Middle	1.25E+04	2.62E+04	4.15E-17
	Average			4.21E-17
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Middle	2.90E+04	3.69E+05	5.28E-17
	Average			5.28E-17
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Middle	8.71E+02	3.27E+03	4.98E-19
	Average			4.98E-19
$^{238}\text{U} (n,f) ^{95}\text{Zr} (\text{Cd})$	Middle	1.49E+03	2.58E+04	2.25E-16
	Including γ fission corrections:			2.18E-16
$^{238}\text{U} (n,f) ^{103}\text{Ru} (\text{Cd})$	Middle	3.01E+02	2.99E+04	2.15E-16
	Including γ fission corrections:			2.08E-16
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	1.69E+03	3.05E+04	2.27E-16
	Including γ fission corrections:			2.20E-16
$^{237}\text{Np} (n,f) ^{95}\text{Zr} (\text{Cd})$	Middle	2.59E+04	4.48E+05	3.55E-15
	Including γ fission corrections:			3.53E-15
$^{237}\text{Np} (n,f) ^{103}\text{Ru} (\text{Cd})$	Middle	4.83E+03	4.80E+05	3.89E-15
	Including γ fission corrections:			3.86E-15
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	2.72E+04	4.90E+05	3.58E-15
	Including γ fission corrections:			3.56E-15
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Middle	9.12E+05	3.42E+06	6.93E-14
	Average			6.93E-14
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Middle	4.70E+05	1.76E+06	3.57E-14
	Average			3.57E-14
Notes:				
1. Measured specific activities are indexed to a counting date of July 24, 2007.				
2. The average $^{238}\text{U} (n,f)$ reaction rates include correction factors to account for plutonium build-in and photo-fission effects in the sensor.				
3. The average $^{237}\text{Np} (n,f)$ reaction rates include correction factors to account for photo-fission effects in the sensor.				

Table C-4 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 EVND Capsule D (cont.)				
Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{46}\text{Ti} (n,p) ^{46}\text{Sc}$	Top	2.50E+02	2.13E+03	2.05E-18
	Average			2.05E-18
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Top	3.54E+03	7.33E+03	1.16E-17
	Top	3.74E+03	7.74E+03	1.23E-17
	Average			1.20E-17
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Top	9.58E+03	1.19E+05	1.71E-17
	Average			1.71E-17
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Top	2.24E+02	8.39E+02	1.28E-19
	Average			1.28E-19
$^{238}\text{U} (n,f) ^{95}\text{Zr} (\text{Cd})$	Top	6.01E+02	1.02E+04	8.90E-17
	Including γ fission corrections:			8.58E-17
$^{238}\text{U} (n,f) ^{103}\text{Ru} (\text{Cd})$	Top	1.18E+02	1.15E+04	8.25E-17
	Including γ fission corrections:			7.95E-17
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Top	5.71E+02	1.03E+04	7.68E-17
	Including γ fission corrections:			7.40E-17
$^{237}\text{Np} (n,f) ^{95}\text{Zr} (\text{Cd})$	Top	1.13E+04	1.92E+05	1.52E-15
	Including γ fission corrections:			1.51E-15
$^{237}\text{Np} (n,f) ^{103}\text{Ru} (\text{Cd})$	Top	2.15E+03	2.09E+05	1.70E-15
	Including γ fission corrections:			1.69E-15
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Top	1.08E+04	1.95E+05	1.42E-15
	Including γ fission corrections:			1.41E-15
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Top	2.43E+05	9.10E+05	1.84E-14
	Average			1.84E-14
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Top	1.57E+05	5.88E+05	1.19E-14
	Average			1.19E-14
Notes:				
1. Measured specific activities are indexed to a counting date of July 24, 2007.				
2. The average $^{238}\text{U} (n,f)$ reaction rates include correction factors to account for plutonium build-in and photo-fission effects in the sensor.				
3. The average $^{237}\text{Np} (n,f)$ reaction rates include correction factors to account for photo-fission effects in the sensor.				

Table C-4 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 EVND Capsule E (cont.)				
Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{46}\text{Ti} (n,p) ^{46}\text{Sc}$	Middle	5.99E+02	4.98E+03	4.80E-18
	Average			4.80E-18
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Middle	9.23E+03	1.89E+04	2.99E-17
	Middle	9.29E+03	1.90E+04	3.01E-17
	Average			3.00E-17
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Middle	2.21E+04	2.69E+05	3.85E-17
	Average			3.85E-17
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Middle	5.84E+02	2.18E+03	3.33E-19
	Average			3.33E-19
$^{238}\text{U} (n,f) ^{95}\text{Zr} (\text{Cd})$	Middle	1.27E+03	2.10E+04	1.84E-16
	Including γ fission corrections:			1.78E-16
$^{238}\text{U} (n,f) ^{103}\text{Ru} (\text{Cd})$	Middle	2.59E+02	2.46E+04	1.77E-16
	Including γ fission corrections:			1.71E-16
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	1.32E+03	2.38E+04	1.78E-16
	Including γ fission corrections:			1.72E-16
$^{237}\text{Np} (n,f) ^{95}\text{Zr} (\text{Cd})$	Middle	2.48E+04	4.11E+05	3.25E-15
	Including γ fission corrections:			3.23E-15
$^{237}\text{Np} (n,f) ^{103}\text{Ru} (\text{Cd})$	Middle	4.18E+03	3.97E+05	3.22E-15
	Including γ fission corrections:			3.20E-15
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Middle	2.45E+04	4.41E+05	3.22E-15
	Including γ fission corrections:			3.21E-15
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Middle	6.17E+05	2.30E+06	4.67E-14
	Average			4.67E-14
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Middle	3.62E+05	1.35E+06	2.74E-14
	Average			2.74E-14
Notes:				
1. Measured specific activities are indexed to a counting date of July 24, 2007.				
2. The average $^{238}\text{U} (n,f)$ reaction rates include correction factors to account for plutonium build-in and photo-fission effects in the sensor.				
3. The average $^{237}\text{Np} (n,f)$ reaction rates include correction factors to account for photo-fission effects in the sensor.				

Table C-4 Measured Sensor Activities and Reaction Rates for Catawba Unit 1 EVND Capsule F (cont.)				
Reaction	Location	Measured Activity (dps/g)	Saturated Activity (dps/g)	Adjusted Reaction Rate (rps/atom)
$^{46}\text{Ti} (n,p) ^{46}\text{Sc}$	Bottom	2.01E+02	1.69E+03	1.63E-18
	Average			1.63E-18
$^{54}\text{Fe} (n,p) ^{54}\text{Mn}$	Bottom	3.06E+03	6.29E+03	9.97E-18
	Bottom	2.88E+03	5.92E+03	9.38E-18
	Average			9.68E-18
$^{58}\text{Ni} (n,p) ^{58}\text{Co}$	Bottom	8.07E+03	9.92E+04	1.42E-17
	Average			1.42E-17
$^{63}\text{Cu} (n,\alpha) ^{60}\text{Co}$	Bottom	1.83E+02	6.84E+02	1.04E-19
	Average			1.04E-19
$^{238}\text{U} (n,f) ^{95}\text{Zr} (\text{Cd})$	Bottom	4.05E+02	6.78E+03	5.92E-17
	Including γ fission corrections:			5.71E-17
$^{238}\text{U} (n,f) ^{103}\text{Ru} (\text{Cd})$	Bottom	8.56E+01	8.22E+03	5.90E-17
	Including γ fission corrections:			5.70E-17
$^{238}\text{U} (n,f) ^{137}\text{Cs} (\text{Cd})$	Bottom	3.86E+02	6.95E+03	5.19E-17
	Including γ fission corrections:			5.01E-17
$^{237}\text{Np} (n,f) ^{95}\text{Zr} (\text{Cd})$	Bottom	8.14E+03	1.36E+05	1.08E-15
	Including γ fission corrections:			1.07E-15
$^{237}\text{Np} (n,f) ^{103}\text{Ru} (\text{Cd})$	Bottom	1.36E+03	1.31E+05	1.06E-15
	Including γ fission corrections:			1.05E-15
$^{237}\text{Np} (n,f) ^{137}\text{Cs} (\text{Cd})$	Bottom	7.11E+03	1.28E+05	9.36E-16
	Including γ fission corrections:			9.30E-16
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$	Bottom	3.14E+05	1.17E+06	2.38E-14
	Average			2.38E-14
$^{59}\text{Co} (n,\gamma) ^{60}\text{Co} (\text{Cd})$	Bottom	1.65E+05	6.17E+05	1.25E-14
	Average			1.25E-14
Notes:				
1. Measured specific activities are indexed to a counting date of July 24, 2007.				
2. The average $^{238}\text{U} (n,f)$ reaction rates include correction factors to account for plutonium build-in and photo-fission effects in the sensor.				
3. The average $^{237}\text{Np} (n,f)$ reaction rates include correction factors to account for photo-fission effects in the sensor.				

Table C-5 Comparison of M/C and BE Reaction Rates at the Surveillance Capsule Center					
Catawba Unit 1 Capsule Z					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	7.66E-17	6.11E-17	7.16E-17	1.25	1.08
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	7.20E-15	6.93E-15	7.33E-15	1.04	0.98
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	9.01E-15	9.71E-15	9.93E-15	0.93	0.91
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	6.02E-12	5.16E-12	5.94E-12	1.17	1.01
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	2.81E-12	3.53E-12	2.86E-12	0.80	0.98
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	4.47E-14	3.72E-14	3.90E-14	1.20	1.15
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	4.25E-13	3.58E-13	3.99E-13	1.19	1.06
Catawba Unit 1 Capsule Y					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	4.98E-17	4.67E-17	4.77E-17	1.07	1.04
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	4.58E-15	5.12E-15	4.83E-15	0.89	0.95
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	6.74E-15	7.15E-15	6.81E-15	0.94	0.99
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	3.56E-12	3.71E-12	3.53E-12	0.96	1.01
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	2.06E-12	2.57E-12	2.09E-12	0.80	0.99
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	2.69E-14	2.70E-14	2.56E-14	1.00	1.05
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	2.57E-13	2.55E-13	2.49E-13	1.01	1.03
Catawba Unit 1 Capsule V					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	4.25E-17	4.43E-17	4.11E-17	0.96	1.03
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	4.17E-15	4.84E-15	4.29E-15	0.86	0.97
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	5.62E-15	6.76E-15	5.93E-15	0.83	0.95
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	3.14E-12	3.47E-12	3.11E-12	0.91	1.01
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	1.69E-12	2.41E-12	1.72E-12	0.7	0.98
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	2.72E-14	2.55E-14	2.30E-14	1.07	1.18
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	2.21E-13	2.40E-13	2.20E-13	0.92	1.00
Note:					
See Section C.1.2 for details describing the BE reaction rates.					

Table C-5 Comparison of M/C and BE Reaction Rates at the Surveillance Capsule Center (cont.)

Catawba Unit 1 Capsule U					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	4.50E-17	4.57E-17	4.33E-17	0.99	1.04
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	4.33E-15	5.02E-15	4.49E-15	0.86	0.96
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	5.93E-15	7.01E-15	6.23E-15	0.85	0.95
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	3.15E-12	3.67E-12	3.13E-12	0.86	1.01
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	1.78E-12	2.53E-12	1.81E-12	0.70	0.98
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	2.89E-14	2.65E-14	2.41E-14	1.09	1.20
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	2.24E-13	2.51E-13	2.27E-13	0.89	0.99
Catawba Unit 1 Capsule X					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	4.56E-17	4.57E-17	4.37E-17	1.00	1.04
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	4.35E-15	5.02E-15	4.61E-15	0.87	0.94
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	6.07E-15	7.01E-15	6.45E-15	0.87	0.94
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	3.44E-12	3.67E-12	3.41E-12	0.94	1.01
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	1.84E-12	2.53E-12	1.88E-12	0.73	0.98
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	3.19E-14	2.65E-14	2.58E-14	1.20	1.23
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	2.94E-13	2.51E-13	2.75E-13	1.17	1.08
Note: See Section C.1.2 for details describing the BE reaction rates.					

Table C-6 Comparison of Calculated and BE Exposure Rates at the EVND Capsule Center from Catawba Unit 1					
Catawba Unit 1 EVND Capsule A					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	4.88E-18	5.57E-18	4.92E-18	0.88	0.99
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	2.70E-17	2.96E-17	2.66E-17	0.91	1.02
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	3.63E-17	4.11E-17	3.67E-17	0.88	0.99
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	3.64E-19	4.12E-19	3.64E-19	0.88	1.00
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	3.99E-14	7.54E-14	4.06E-14	0.53	0.98
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	1.87E-14	2.25E-14	1.87E-14	0.83	1.00
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	1.24E-16	1.44E-16	1.30E-16	0.86	0.95
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	1.97E-15	1.92E-15	1.87E-15	1.03	1.05
Catawba Unit 1 EVND Capsule B					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	6.92E-18	7.38E-18	6.86E-18	0.94	1.01
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	3.93E-17	4.09E-17	3.87E-17	0.96	1.02
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	5.31E-17	5.71E-17	5.38E-17	0.93	0.99
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	4.77E-19	5.33E-19	4.85E-19	0.90	0.98
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	5.80E-14	1.03E-13	5.90E-14	0.56	0.98
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	2.99E-14	3.61E-14	2.98E-14	0.83	1.00
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	1.94E-16	2.06E-16	1.99E-16	0.94	0.98
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	3.17E-15	2.87E-15	3.00E-15	1.10	1.05
Catawba Unit 1 EVND Capsule C					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	7.00E-18	7.75E-18	7.00E-18	0.90	1.00
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	4.21E-17	4.39E-17	4.05E-17	0.96	1.04
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	5.28E-17	6.18E-17	5.56E-17	0.85	0.95
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	4.98E-19	5.55E-19	4.99E-19	0.90	1.00
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	6.93E-14	1.22E-13	7.05E-14	0.57	0.98
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	3.57E-14	4.31E-14	3.56E-14	0.83	1.00
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	2.15E-16	2.30E-16	2.15E-16	0.93	1.00
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	3.65E-15	3.37E-15	3.43E-15	1.08	1.06
Note: See Section C.1.2 for details describing the BE reaction rates.					

Table C-6 Comparison of Calculated and BE Exposure Rates at the EVND Capsule Center from Catawba Unit 1 (cont.)					
Catawba Unit 1 EVND Capsule D					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	2.05E-18	2.19E-18	1.98E-18	0.94	1.04
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	1.19E-17	1.31E-17	1.21E-17	0.91	0.99
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	1.71E-17	1.88E-17	1.75E-17	0.91	0.98
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	1.28E-19	1.55E-19	1.32E-19	0.83	0.97
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	1.84E-14	5.20E-14	1.93E-14	0.35	0.95
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	1.19E-14	1.54E-14	1.17E-14	0.78	1.02
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	7.98E-17	7.55E-17	7.48E-17	1.06	1.06
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	1.54E-15	1.19E-15	1.39E-15	1.29	1.11
Catawba Unit 1 EVND Capsule E					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	4.79E-18	5.54E-18	4.76E-18	0.87	1.01
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	3.00E-17	3.32E-17	2.90E-17	0.90	1.03
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	3.85E-17	4.76E-17	4.07E-17	0.81	0.94
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	3.33E-19	3.92E-19	3.34E-19	0.85	1.00
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	4.67E-14	1.11E-13	4.81E-14	0.42	0.97
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	2.74E-14	3.47E-14	2.71E-14	0.79	1.01
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	1.74E-16	1.92E-16	1.71E-16	0.91	1.01
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	3.21E-15	3.00E-15	2.99E-15	1.07	1.08
Catawba Unit 1 EVND Capsule F					
Reaction	Reaction Rate [rps/atom]			M/C	M/BE
	Measured	Calculated	BE		
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	1.63E-18	2.13E-18	1.59E-18	0.76	1.02
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	9.68E-18	1.28E-17	9.67E-18	0.76	1.00
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	1.42E-17	1.84E-17	1.41E-17	0.77	1.01
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	1.04E-19	1.50E-19	1.08E-19	0.69	0.96
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}$	2.38E-14	5.21E-14	2.43E-14	0.46	0.98
$^{59}\text{Co}(n,\gamma)^{60}\text{Co}(\text{Cd})$	1.25E-14	1.54E-14	1.24E-14	0.81	1.01
$^{238}\text{U}(n,f)^{137}\text{Cs}(\text{Cd})$	5.47E-17	7.44E-17	5.79E-17	0.74	0.94
$^{237}\text{Np}(n,f)^{137}\text{Cs}(\text{Cd})$	1.02E-15	1.19E-15	9.89E-16	0.86	1.03
Note:					
See Section C.1.2 for details describing the BE reaction rates.					

Table C-7 Comparison of Calculated and BE Exposure Rates at the Surveillance Capsule Center from Catawba Unit 1

Capsule ID	$\phi(E > 1.0 \text{ MeV}) [\text{n/cm}^2\text{-s}]$			
	Calculated	BE	Uncertainty (1 σ)	BE/C
Z	1.17E+11	1.23E+11	6%	1.04
Y	8.40E+10	7.94E+10	a. 6%	0.94
V	7.91E+10	7.19E+10	6%	0.90
U	8.26E+10	7.52E+10	6%	0.91
X	8.26E+10	8.25E+10	6%	0.99

Note:

Calculated results are based on the RAPTOR-M3G transport calculations taken at the core midplane following the completion of each respective capsules irradiation period and are the average neutron exposure over the irradiation period for each capsule. See Section C.1.2 for details describing the BE exposure rates.

Capsule ID	Iron Atom Displacement Rate [dpa/s]			
	Calculated	BE	Uncertainty (1 σ)	BE/C
Z	2.23E-10	2.38E-10	8%	1.06
Y	1.59E-10	1.53E-10	8%	0.96
V	1.50E-10	1.39E-10	8%	0.92
U	1.57E-10	1.45E-10	8%	0.92
X	1.57E-10	1.61E-10	8%	1.02

Note:

Calculated results are based on the RAPTOR-M3G transport calculations taken at the core midplane following the completion of each respective capsules irradiation period and are the average neutron exposure over the irradiation period for each capsule. See Section C.1.2 for details describing the BE exposure rates.

Table C-8 Comparison of M/C Sensor Reaction Rate Ratios Including all Fast Neutron Threshold Reactions from Catawba Unit 1							
Reaction	M/C Ratio					Average	% Standard Deviation
	Capsule Z	Capsule Y	Capsule V	Capsule U	Capsule X		
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	1.25	1.07	0.96	0.99	1.00	1.05	11.1
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	1.04	0.89	0.86	0.86	0.87	0.90	8.5
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	0.93	0.94	0.83	0.85	0.87	0.88	5.5
$^{238}\text{U}(n,f)^{137}\text{Cs (Cd)}$	1.20	1.00	1.07	1.09	1.20	1.11	7.8
$^{237}\text{Np}(n,f)^{137}\text{Cs (Cd)}$	1.19	1.01	0.92	0.89	1.17	1.04	13.4
Average	1.12	0.98	0.93	0.94	1.02	1.00	
% Standard Deviation	11.8	7.0	10.1	10.9	15.5		12.8

Note: The average and % standard deviation values in bold face type represent the average and standard deviation of the entire twenty-five sample threshold foil data set.

Table C-9 Comparison of BE/C Exposure Rate Ratios for Surveillance Capsules from Catawba Unit 1		
Capsule ID	BE/C Ratio	
	$\phi(E > 1.0 \text{ MeV})$	dpa/s
Z	1.04	1.06
Y	0.94	0.96
V	0.90	0.92
U	0.91	0.92
X	0.99	1.02
Average	0.96	0.98
% Standard Deviation	6.1%	6.4%

Table C-10 Comparison of Calculated and BE Exposure Rates at the EVND Capsules at the Core Midplane of Catawba Unit 1

Capsule ID	$\phi(E > 1.0 \text{ MeV}) [\text{n/cm}^2\text{-s}]$			
	Calculated	BE	Uncertainty (1σ)	BE/C
A	4.62E+08	4.24E+08	6%	0.91
B	6.77E+08	6.64E+08	6%	0.98
C	7.73E+08	7.33E+08	6%	0.94
E	6.70E+08	6.09E+08	6%	0.90
Note: Calculated results are based on the RAPTOR-M3G transport calculations taken at the core midplane following the completion of each respective capsules irradiation period and are the average neutron exposure over the irradiation period for each capsule. See Section C.1.2 for details describing the BE exposure rates.				
Capsule ID	Iron Atom Displacement Rate [dpa/s]			
	Calculated	BE	Uncertainty (1σ)	BE/C
A	1.43E-12	1.33E-12	9%	0.92
B	2.17E-12	2.14E-12	9%	0.98
C	2.56E-12	2.46E-12	9%	0.96
E	2.24E-12	2.08E-12	9%	0.92
Note: Calculated results are based on the RAPTOR-M3G transport calculations taken at the core midplane following the completion of each respective capsules irradiation period and are the average neutron exposure over the irradiation period for each capsule. See Section C.1.2 for details describing the BE exposure rates.				

Table C-11 Comparison of Calculated and BE Exposure Rates at the EVND Capsules at Off-Midplane Positions of Catawba Unit 1				
Capsule ID	$\phi(E > 1.0 \text{ MeV}) [\text{n/cm}^2\text{-s}]$			
	Calculated	BE	Uncertainty (1σ)	BE/C
D	2.63E+08	2.70E+08	6%	1.02
F	2.60E+08	2.06E+08	6%	0.79
Note: Calculated results are based on the RAPTOR-M3G transport calculations taken at the EVND capsule center following the completion of each respective capsules irradiation period and are the average neutron exposure over the irradiation period for each capsule. See Section C.1.2 for details describing the BE exposure rates.				
Capsule ID	Iron Atom Displacement Rate [dpa/s]			
	Calculated	BE	Uncertainty (1σ)	BE/C
D	8.99E-13	9.34E-13	9%	1.03
F	8.97E-13	7.28E-13	9%	0.81
Note: Calculated results are based on the RAPTOR-M3G transport calculations taken at the EVND capsule center following the completion of each respective capsules irradiation period and are the average neutron exposure over the irradiation period for each capsule. See Section C.1.2 for details describing the BE exposure rates.				

Table C-12 Comparison of M/C Sensor Threshold Reaction Rates from Ex-Vessel Midplane Capsules at Catawba Unit 1

Reaction	M/C Ratio				Average	% Standard Deviation
	Capsule A	Capsule B	Capsule C	Capsule E		
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	0.88	0.94	0.90	0.87	0.90	3.4
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	0.91	0.96	0.96	0.90	0.93	3.4
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	0.88	0.93	0.85	0.81	0.87	5.8
$^{63}\text{Cu}(n,a)^{60}\text{Co}$	0.88	0.90	0.90	0.85	0.88	2.7
$^{238}\text{U}(n,f)^{137}\text{Cs (Cd)}$	0.86	0.94	0.93	0.91	0.91	3.9
$^{237}\text{Np}(n,f)^{137}\text{Cs (Cd)}$	1.03	1.10	1.08	1.07	1.07	2.8
Average	0.91	0.96	0.94	0.90	0.93	
% Standard Deviation	6.9	7.3	8.5	10.0		8.1

Note: The average and % standard deviation values in bold face type represent the average and standard deviation of the entire twenty-four sample threshold foil data set.

Table C-13 Comparison of M/C Sensor Threshold Reaction Rates from Ex-Vessel Off-Midplane Capsules at Catawba Unit 1				
Reaction	M/C Ratio		Average	% Standard Deviation
	Capsule D	Capsule F		
$^{46}\text{Ti}(n,p)^{46}\text{Sc}$	0.94	0.76	0.85	15.0
$^{54}\text{Fe}(n,p)^{54}\text{Mn}$	0.91	0.76	0.84	12.7
$^{58}\text{Ni}(n,p)^{58}\text{Co}$	0.91	0.77	0.84	11.8
$^{63}\text{Cu}(n,\alpha)^{60}\text{Co}$	0.83	0.69	0.76	13.0
$^{238}\text{U}(n,f)^{137}\text{Cs (Cd)}$	1.06	0.74	0.90	25.1
$^{237}\text{Np}(n,f)^{137}\text{Cs (Cd)}$	1.29	0.86	1.08	28.3
Average	0.99	0.76	0.88	
% Standard Deviation	16.6	7.3		19.0

Note: The average and % standard deviation values in bold face type represent the average and standard deviation of the entire twelve sample threshold foil data set.

Table C-14 Comparison of BE/C Exposure Rate Ratios from Ex-Vessel Midplane Capsules at Catawba Unit 1		
Capsule ID	BE/C Ratio	
	$\phi(E > 1.0 \text{ MeV})$	dpa/s
A	0.91	0.92
B	0.98	0.98
C	0.94	0.96
E	0.90	0.92
Average	0.93	0.95
% Standard Deviation	3.9%	3.2%

Table C-15 Comparison of BE/C Exposure Rate Ratios from Ex-Vessel Off-Midplane Capsules at Catawba Unit 1		
Capsule ID	BE/C Ratio	
	$\phi(E > 1.0 \text{ MeV})$	dpa/s
D	1.02	1.03
F	0.79	0.81
Average	0.91	0.92
% Standard Deviation	18.0%	16.9%

C.2 REFERENCES

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